Seat Majorities without Vote Majorities: Coalition Inversions in Brazil's Chamber of Deputies

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Abstract

Proportional representation is usually evaluated party by party, by comparing each party's share of seats with its share of votes. Yet fragmented legislatures are governed through combinations of parties, so small party-level deviations can become consequential when parties are combined. Brazil provides a useful setting because the Chamber of Deputies is assembled from twenty-seven separate district elections. Several institutional steps separate national votes from final party seat totals, and no national compensatory allocation reconciles them. I ask whether politically meaningful sets of parties can hold a legislative majority despite receiving less than half of all valid federal-deputy votes nationwide, a condition known as coalition inversion. Using the 2014, 2018, and 2022 elections, I compare distinct election-year cabinet party sets with potential coalitions restricted by ideological proximity. Cabinet inversions appear following the 2014 and 2022 elections, but not the 2018 election; inversions among minimal connected winning coalitions appear in all three. In 2022, a seven-party bloc of ideologically adjacent parties combined 45.26 percent of the vote with 258 of 513 seats. Party-level deviations from proportionality can therefore have a distinct coalition-level consequence: they can determine which combinations of parties command a legislative majority.

1 Introduction

Electoral systems translate votes into seats. Under proportional representation, the quality of that translation is usually assessed by comparing each party's share of the vote with its share of the seats (Kaminski 2026). The comparison is indispensable, but it stops at the level of the units that contest the election. In fragmented legislatures, no single party need command a majority, and the majorities that organize government and legislative action are assembled from several parties.

Combining parties changes the significance of their individual deviations from proportionality. One party may receive slightly more seats than its national vote share implies, while another receives slightly fewer. When parties are grouped together, these deviations can cancel or reinforce one another. A set of parties may therefore cross the legislative majority threshold before its combined vote share reaches one half, leaving the complementary set with more votes but fewer seats. This configuration is a coalition inversion: a set of parties holds a seat majority while receiving less than half of all valid national votes cast for the legislature.

Kurrild-Klitgaard (2013) documents empirical cases of coalition inversions in Danish government formation. Miller (2014) establishes the more general result that inversions remain possible even when seats are allocated in a single national constituency, without an explicit seat threshold, and by a highly proportional formula. Because seats must be assigned in whole numbers, party seat shares will generally depart slightly from exact proportionality. Once positive and negative party-level deviations exist, some combinations of parties can reverse the vote and seat majorities. These results settle the existence question. They do not show how coalition inversions appear when national party seat totals are themselves assembled from many district elections.

Brazil provides such a setting. The Chamber of Deputies is not formed by applying one proportional allocation to national party votes. Its seats are allocated separately across the twenty-six states and the Federal District. District magnitudes and electoral weights vary, electoral rules and the units over which votes were pooled changed across the three elections, and party support is distributed unevenly across the country. Several institutional and geographical mediations therefore stand between a party's national vote share and its final national seat share (Nicolau 2006; Samuels and Snyder 2001; Bochsler 2026).

This institutional structure changes the sources of deviation without changing the national proportionality benchmark. I compare seat shares with shares of all valid federal-deputy votes, including votes for parties that obtain no seats. Restricting the denominator to votes for seat-winning parties would instead evaluate representation conditional on which parties entered the Chamber. The question here concerns the national outcome of the entire vote-to-seat process, including electoral exclusion. The benchmark therefore measures departures from national proportionality without presuming that those departures are unintended or unjustified.

Party-level deviations emerge from twenty-seven district-level translations operating under heterogeneous conditions, rather than only from rounding a single national allocation. The organizing question is therefore: when do these deviations combine so that a politically meaningful set of parties holds a legislative majority without a national vote majority? The contribution lies in showing how the resulting party-level seat differentials aggregate across coalitions and change majority status.

The Chamber contains an enormous number of mathematically possible party subsets, most of which have little political meaning. Realized cabinet coalitions provide the first empirical domain. Brazilian presidents normally distribute cabinet offices among several parties, and changes in ministerial participation create successive party sets during a legislative term (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). Each change may produce a new realized party set. Comparing that set's Chamber seats with the votes received by its members in the corresponding election shows when a cabinet coalition held a legislative majority without a national vote majority.

Cabinets are politically realized, but they are also selective and may combine parties separated by substantial ideological distance. A second domain asks whether the same aggregation problem appears among potential coalitions disciplined by ideological proximity. The baseline uses minimal connected winning coalitions: contiguous blocs in the election-year left-right order that hold a seat majority and cannot be narrowed further while retaining it (Axelrod 1970). Exact connectedness is deliberately restrictive; a nested one-gap robustness check permits one represented party inside a coalition's ideological span to be omitted. For cabinets and potential coalitions, the relevant quantity is the coalition's accumulated seat differential, defined as the difference between its seats and the number implied by its national vote share. That differential admits an exact algebraic decomposition into a within-district term and a between-district term. The terms make cases comparable by showing whether the two components reinforce or offset one another.

Inversions appear in both coalition domains, but their distribution differs across elections. Two of the 35 distinct cabinet party sets invert: one observed for five days under Dilma Rousseff following the 2014 election and one observed during the first 255 days of Lula III following the

2022 election. No cabinet party set linked to the 2018 election inverts in the primary chronology. Minimal connected winning inversions, by contrast, appear in all three elections. The 2022 cases include a compact right-side bloc with 45.26 percent of the vote and 258 seats. Within-district and between-district components reinforce one another in some inversions and offset one another in others.

The remainder of the article follows the order of these transformations. Section 2 explains how Brazil's multidistrict open-list system converts district votes into the national party seat totals that coalitions later combine. Section 3 defines coalition inversion and the comparative decomposition, and Section 3.2 describes the data. Sections 4–4.3 move from party-level deviations to realized cabinet coalitions, ideologically constrained coalitions, and the comparison of decomposition patterns. Section 5 discusses the implications, and the final section concludes.

2 Brazil's multidistrict open-list proportional system

Before coalition majorities can be compared, one institutional question must be answered: how do twenty-seven separate electoral contests produce the national party seat distribution that coalitions later combine? Elections to Brazil's Chamber of Deputies take place inside the states, not in a single national constituency. Each of the twenty-six states and the Federal District forms a separate electoral circumscription for federal-deputy elections[1]. Seats are allocated independently in these twenty-seven districts and together compose the 513-member Chamber. No national compensatory calculation revises the party totals once the district counts have been completed (Brasil 1993; Câmara dos Deputados 2026; Tribunal Superior Eleitoral 2021b).

The number of seats at stake varies sharply across districts. The constitutional floor is eight and the ceiling is seventy. Eleven districts elect eight deputies, the median magnitude is ten, the mean is nineteen, and São Paulo alone elects seventy. This distribution matters because the mechanical environment faced by parties in most districts is much smaller than the national size of the Chamber suggests. Much of Brazilian proportional representation operates near the lower end of the district-magnitude range, where each seat is a relatively large indivisible share of the delegation and the effective barrier to representation tends to be higher. Larger districts offer a more permissive environment, but because seats are allocated independently across states, district-level representational shortfalls are not compensated at the national level.

Within each district, voters encounter an open-list election. They may enter the identifying number of an individual candidate or cast a vote for a party label. A candidate vote performs two functions at once. It contributes to the aggregate vote of the party, electoral coalition, or federation through which the candidate competes, and it raises that candidate's position within the relevant open list. A party-label vote contributes to the first calculation but not to the ranking among candidates. That is, votes first determine how many seats an electoral list receives. Nominal votes then determine which candidates within that list take the seats, subject to the individual eligibility requirements in force for the election. A candidate can therefore benefit from votes cast for copartisans or electoral partners, but cannot ordinarily occupy a seat unless the list to which the candidate belongs wins one (Nicolau 2006; Brasil 1997).

The count begins by aggregating all valid candidate and party-label votes credited to the relevant electoral unit in district d. Depending on the election, that unit may be a party, a proportional electoral coalition, or a party federation. The institutional differences among these arrangements are discussed below. The electoral quotient, conventionally denoted by QE, is obtained by dividing

1. The Federal District is therefore one ordinary eight-seat district within this architecture.

the district's valid proportional-election vote by its magnitude and applying the statutory rounding rule. The party quotient then divides the vote total of each electoral unit by QE and discards the fractional part:

$$QE_d = \text{round}_{\text{stat}}\left(\frac{V_d}{M_d}\right), \qquad QP_{\ell d} = \left\lfloor \frac{v_{\ell d}}{QE_d} \right\rfloor.$$

Here V_d is the total valid vote in district d, M_d is its district magnitude, and $v_{\ell d}$ is the vote credited to electoral unit ℓ. The party quotient determines the unit's initial seat allocation, subject to the candidate-level eligibility requirements applicable in that election (Brasil 1965, arts. 106–108).

Because the party quotient discards fractional remainders, this first stage normally leaves seats unallocated. The remaining seats are distributed sequentially through the statutory highest-averages procedure. If electoral unit ℓ has already been allocated $s_{\ell d}$ seats, its value for the next seat is

$$m_{\ell d} = \frac{v_{\ell d}}{s_{\ell d} + 1}.$$

The denominator includes the seat currently being allocated, so $m_{\ell d}$ is the unit's average number of votes per seat if it were to receive that additional seat. The seat goes to the eligible unit with the highest value, its seat total is updated, and the comparison is repeated until all seats in the district have been allocated. The resulting priority quotients follow the Jefferson–D'Hondt sequence $v_{\ell d}/1, v_{\ell d}/2, v_{\ell d}/3, \ldots$. The D'Hondt highest-averages sequence governs the allocation of remaining seats, but it operates after the electoral- and party-quotient stage and subject to the election-specific eligibility rules described below (Brasil 1965; Nicolau and Schmitt 1995, art. 109).

The elections differed in the legal thresholds governing access to seats and the remainder allocation. In 2014, there was no fixed candidate-level threshold, but only parties or proportional coalitions that reached the electoral quotient could compete for remaining seats (Tribunal Superior Eleitoral 2013, art. 186, §2). In 2018, candidates ordinarily needed at least 10 percent of the electoral quotient to occupy a seat, while all parties and coalitions could enter the remainder competition regardless of whether they had reached the quotient, provided an eligible candidate remained; once no such candidate remained, the final seats were allocated by highest averages without that nominal-vote minimum (Tribunal Superior Eleitoral 2017, arts. 7–10). Under the rules applied in the original 2022 allocation, the 10 percent candidate threshold remained for initial seats, while the first remainder stage required parties or federations to reach 80 percent of the electoral quotient and candidates to reach 20 percent; the final highest-averages stage retained the party-level requirement but not the 20 percent candidate threshold (Tribunal Superior Eleitoral 2021b, arts. 8–11). These changes affected eligibility for marginal seats rather than the basic allocation sequence described above.

Moreover, proportional electoral coalitions, or *coligações proporcionais*, were part of the 2014 and 2018 elections. They were formed within the electoral circumscription, which for the Chamber meant a state or the Federal District. A party could consequently enter different proportional arrangements in different states, subject to the legal restrictions in force for the election. These were not national alliances whose composition automatically carried across all twenty-seven contests (Nicolau 2006; Tribunal Superior Eleitoral 2013, 2017). Member parties of proportional electoral coalitions legally operated as a joint open list when votes were converted into seats. Candidate votes and party-label votes credited to any member entered the coalition's district total; the quotient and highest-averages calculations then assigned seats to the coalition as a whole. Those seats were occupied by the highest nominal vote-getters across the combined list, regardless of the member party to which they belonged. A candidate from one party could therefore be elected partly because votes cast for candidates or labels of another coalition member raised the joint total

(Nicolau 2006; Tribunal Superior Eleitoral 2013, 2017). However, pooling did not make the parties disappear. Candidates remained affiliated with member parties, used party identifiers on the ballot, and campaigned in a setting where party labels and candidate numbers remained central. The coalition changed the pool within which votes were translated into seats; it did not convert its members into a single political organization. Nor did it promise a stable proportional return to each partner. The number of seats won by the joint list was collective, while the distribution of those seats among member parties depended on how their candidates ranked in nominal votes across the coalition.

This arrangement has a clear family resemblance to list apparentement because several parties combine votes before the seat total is settled. The canonical apparentement studied by Pukelsheim and Leutgäb (2013), however, is a two-level apportionment: the alliance first receives seats collectively, after which those seats are apportioned among its component lists. Brazil instead used a single joint open list, with candidate nominal votes selecting the winners across party boundaries. Vote pooling can improve the electoral position of allied lists, while the allocation of the resulting advantage among particular partners is highly sensitive to the configuration of votes and alliances and can acquire a lottery-like quality. In Brazil, too, vote pooling could improve the electoral prospects of an alliance or of particular member parties, but any member-party gains depended on the distribution of candidate votes within the joint list rather than on a fixed rule allocating bonus seats to coalition partners.

Constitutional reform ended proportional electoral coalitions before the 2022 general election, and party federations became the available form of multiparty pooling in proportional contests (Brasil 2017, 1995). Three federations competed that year: Federação Brasil da Esperança, joining PT, PCdoB, and PV; Federação PSDB Cidadania; and Federação PSOL REDE (Tribunal Superior Eleitoral 2026). Like the former coligações, a federation pooled its members' votes to determine how many seats its common list received, while nominal votes determined which candidates occupied those seats. The resulting seat totals of the constituent parties were therefore not independent proportional allocations from their own vote totals, but ex post outcomes of competition within the pooled list.

The principal difference from proportional electoral coalitions lies in the federation's durability and organizational scope. A federation operates nationally and, in principle, persists for at least four years, whereas the former proportional coalitions were temporary electoral arrangements. This persistence carries some procedural consequences inside Congress, including common representation for certain leadership and committee-allocation purposes, but it does not merge the constituent parties: they retain separate identities and organizational autonomy (Brasil 1995; Tribunal Superior Eleitoral 2021a, 2022). Federations therefore add another institutional mediation between party votes and the party-level seat totals ultimately observed in the Chamber, rather than replacing those parties as political actors. (Brasil 1995; Tribunal Superior Eleitoral 2021a, 2022, 2019).

National proportionality provides a common baseline against which to assess how Brazil's layered vote-to-seat process generates representational deviations and when those deviations become majority-changing as parties are recombined into coalitions. If v_i is the national valid vote total credited to party i and V is the corresponding total for all parties, including those that win no seats, the party's reference quota is $513v_i/V$ seats. Brazil's actual system never performs this national allocation.

Several mediations therefore stand between a party's national vote share and its national seat share. Within each district, finite magnitude, the quotient and highest-averages calculations, legal eligibility thresholds, and the number and relative strength of competing electoral units shape the conversion of votes into seats. Electoral pooling adds a further mediation by changing both the vote total on which seats are allocated and, through the common candidate list, the eventual distribution of those seats among member parties. These arrangements varied across districts in 2014 and 2018 and became nationally fixed through federations in 2022.

The national seat distribution then aggregates twenty-seven independently determined district outcomes without a compensatory tier. Chamber seats are apportioned by population, but the constitutional floor of eight and ceiling of seventy prevent exact population-to-seat equality across states (Samuels and Snyder 2001; Brasil 1993). Turnout and valid-vote rates also differ across elections and districts, so a seat can correspond to different numbers of valid votes in different states. Because parties distribute their support unevenly across the country, national vote shares can therefore be assembled from geographically different configurations with different representational consequences. Electoral geography interacts with district seat weights and local competitive conditions before the resulting party seats are summed nationally. The resulting disproportionality need not have a single source: allocation rules, district magnitude, malapportionment and electoral weight, electoral geography, and party competition constitute distinct channels through which party vote and seat shares can diverge (Bochsler 2026).

This layered translation occurs in an unusually fragmented party environment. Measured by the Laakso–Taagepera index, the effective numbers of electoral parties were 14.11 in 2014, 18.02 in 2018, and 12.34 in 2022; the corresponding effective numbers of parliamentary parties were 13.33, 16.41, and 9.91 (Laakso and Taagepera 1979). The exact number of allocation units was not identical to the number of party labels, especially when district coalitions or national federations pooled votes. Even so, each election produced a Chamber in which legislative majorities had to be assembled across numerous partisan units. Fragmentation makes the political consequences of modest party-level gains and losses depend on how those parties are combined.

The national party seat totals produced by this process enter a presidential system in which governing capacity ordinarily depends on multiparty arrangements. Brazilian coalitional presidentialism links the presidency to combinations of parties across cabinet and Congress: ministries are distributed among several parties, while executive–legislative coordination is organized through agreements among them (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). The Chamber assembled from twenty-seven district elections therefore supplies the parliamentary resources from which governing coalitions are constructed. Party-level representation is politically consequential not only in isolation, but also through the combinations of parties that can command a legislative majority.

National proportionality provides an analytical baseline for examining this second stage. The realized party seat totals incorporate the effects of rounding, highest averages, district magnitude, eligibility requirements, electoral pooling, apportionment, turnout, valid-vote variation, electoral geography, and district-level competition. Comparing those totals with national proportional quotas identifies the positive and negative representational deviations carried by each party without attributing them to any single mechanism. Once parties are combined into politically relevant coalitions, these deviations may reinforce or offset one another. The next section formalizes this aggregation and states the condition under which the accumulated deviation carries a vote-minority coalition across the Chamber's 257-seat majority threshold.

3 Coalition inversions, domains, and measurement

The national seat total received by each party need not coincide with the quantity implied by exact national proportionality. A party may finish with a small positive seat deviation, while another finishes with a small negative one. In isolation, either difference may have little political significance. Coalition formation changes the scale of the comparison because each member brings its deviation into the combined party set. Positive and negative deviations may cancel, but they may also reinforce one another. A set of parties can therefore cross the Chamber's 257-seat majority threshold before its combined vote reaches one half of the national total. The complementary set then has more votes but fewer seats. This is the basic mechanism of coalition inversion.

The possibility of coalition inversions under proportional representation is already established by Kurrild-Klitgaard (Kurrild-Klitgaard 2013) and Miller (Miller 2014). The present analysis extends this agenda to a distinct institutional setting, where national party seat totals emerge from twenty-seven independent district allocations and several additional electoral mediations, and to coalition domains defined by realized cabinet participation and ideological proximity.

A fragmented legislature permits an enormous number of mathematically possible party subsets. An unrestricted search would be exhaustive, but it would place politically remote combinations on the same footing as combinations that were realized in government or structured by recognizable political proximity. The coalition space is therefore restricted in two different ways. Observed cabinets identify realized governing party combinations. Ideological connectedness supplies a disciplined domain of potential coalitions. Taken together, they use coalition inversion as a lens on two different features of the political system: how governing coalitions recombine party-level representational advantages in practice, and where potential vote-minority seat majorities are located within the ideological structure of the legislature.

Cabinet coalitions are substantively meaningful because Brazilian presidents ordinarily govern through multiparty arrangements. Cabinet participation records a party combination that was not merely arithmetically possible but institutionally realized in executive government (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). The relevant representational question is, thus, whether the members of the party set holding ministries collectively possessed a Chamber majority despite receiving less than half of the valid federal-deputy vote in the election that produced that legislature.

This domain must be followed across the legislative term. Coalitional presidentialism is dynamic: parties enter and leave the government, changing the set represented in cabinet while the underlying election-year vote and seat distribution remains fixed. These transitions generate successive realized combinations of the same party-level differentials. Cabinet participation is used only to delimit those observable governing party sets. This provides a useful setting for studying coalition inversion because changes in inversion status can occur without any change in the underlying electoral result: what changes is the particular combination of parties across which the fixed vote-seat differentials are aggregated. The cabinet sequence therefore reveals whether, and for how long, politically realized recombinations of the same legislature cross the majority threshold without a corresponding vote majority.

Observed cabinets reveal which combinations entered government, but they cover only a small portion of the majority configurations permitted by the legislature. A systematic analysis of potential coalitional majorities therefore requires a larger domain. Ideological proximity restricts this combinatorial space and shows whether vote-minority seat majorities can be assembled from ideologically coherent blocs and where such seat-efficient majority regions lie in the party system. The ideological domain, thus, extends the inversion analysis to politically structured potential coalitions. I use Axelrod's connected-coalition criterion as the baseline: an admissible coalition must occupy a contiguous interval in the ideological ordering (Axelrod 1970). The criterion is not intended as a model of Brazilian government formation. It defines a politically structured domain of potential coalitions in which ideological distance constrains which party combinations are compared.

Within this domain, I focus on minimal connected winning coalitions. A connected coalition is minimal winning if it holds at least 257 seats but no proper connected subset also holds a majority. Minimality removes larger intervals that are winning only because they contain a smaller winning coalition and isolates the inclusion frontier at which connected party sets acquire majority status. The inversion question can then be asked directly at that frontier: which of these ideologically coherent seat majorities still fall short of a national vote majority?

3.1 Measurement

Let P denote the set of all parties receiving valid votes in a federal-deputy election, including those that win no seats. For each party $i \in P$, let v_i be its national total of valid candidate and party-label votes and s_i its Chamber seats. With $V = \sum_{i \in P} v_i$ denoting the total valid federal-deputy vote and $S = 513$, exact national proportionality assigns party i the quota

$$q_i = S \frac{v_i}{V}.$$

The party's seat differential is $d_i = s_i - q_i$. A positive differential means that the party receives more seats than its national vote share implies, while a negative differential means that it receives fewer. Since party seats and party quotas both sum to S, the differentials sum to zero. For a coalition $C \subseteq P$, its vote total, seat total, proportional quota, and seat differential are

$$v_C = \sum_{i \in C} v_i, \qquad s_C = \sum_{i \in C} s_i,$$

$$q_C = S \frac{v_C}{V}, \qquad d_C = s_C - q_C.$$

Because coalition votes, seats, and quotas are obtained by summing over members,

$$d_C = s_C - q_C = \sum_{i \in C} (s_i - q_i) = \sum_{i \in C} d_i.$$

The coalition's departure from exact national proportionality is produced by adding the positive and negative differentials of its member parties. A party with a negative contribution can belong to an inverted coalition, provided that the positive contributions elsewhere in the set more than offset it.

A coalition inversion occurs when the party set receives less than half of the national valid federal-deputy vote but controls at least 257 Chamber seats: $\frac{v_C}{V} < \frac{1}{2}$ and $s_C \geq 257$.

For a coalition receiving less than half of the national vote, the seat-majority condition can be written as $d_C \geq 257 - q_C$. The right-hand side is the number of seats separating the coalition's exact proportional quota from a Chamber majority. An inversion occurs when the coalition's accumulated seat differential is large enough to bridge that gap. Since C and its complement $\bar{C} = P \setminus C$ partition both the full election-year party set and the 513 seats, $\frac{v_C}{V} < \frac{1}{2},\ s_C \geq 257 \iff \frac{v_{\bar{C}}}{V} > \frac{1}{2},\ s_{\bar{C}} \leq 256$. The criterion therefore identifies a reversal between the coalition and its complement[2]. Throughout the article, V includes all valid federal-deputy votes rather than

2. This structure differs from Siaroff's (Siaroff 2003) distinction between manufactured and spurious majorities. A manufactured majority occurs when the vote-leading party has less than half of the vote but receives a seat majority; a spurious majority occurs when the seat-majority party received fewer votes than another party. Coalition inversion necessarily has the stronger reversal structure. Because C and its complement $\bar{C}$ exhaust both votes and seats, $v_C/V < 1/2 < s_C/S$ implies that $\bar{C}$ has a vote majority but a seat minority. The outcome is therefore analogous simultaneously to a spurious majority for C and an artificial minority for $\bar{C}$. Siaroff's units, however, are parties or electoral alliances treated as parties, whereas the objects considered here are sets of parties. I therefore retain the term *coalition inversion.*

only votes cast for seat-winning parties or another restricted vote total.

Representation ratios express the same departures in relative rather than absolute terms (Taagepera and Laakso 1980). At the party and coalition levels,

$$R_i = \frac{s_i/S}{v_i/V} = \frac{s_i}{q_i}, \qquad R_C = \frac{s_C/S}{v_C/V} = \frac{s_C}{q_C} = 1 + \frac{d_C}{q_C} = \sum_{i \in C} \frac{q_i}{q_C} R_i.$$

A value of one denotes exact proportionality, a value above one denotes overrepresentation, and a value below one denotes underrepresentation. Parties receiving votes but no seats have $R_i = 0$.The ratios permit comparisons across parties and coalitions of different sizes. The differentials remain the more direct quantities for the majority-threshold question because they measure the relevant departure in seats.

The national coalition differential can be decomposed by adding and subtracting the seats that exact proportionality would assign to the coalition separately within each district:

$$d_C = \left[\sum_d \left(s_{Cd} - S_d \frac{v_{Cd}}{V_d} \right) \right] + \left[\left(\sum_d S_d \frac{v_{Cd}}{V_d} \right) - \left(S \frac{v_C}{V} \right) \right].$$

The first term compares the coalition's actual seats in each district with the number implied by its vote share in that district. I denote it the within-district component, A_C. The other component is the between-district component, B_C. Its first term sums the coalition's proportional quota in each district. The second is the quota implied by the coalition's share of the national valid vote. These quantities need not coincide because districts differ in the number of seats they contribute relative to the number of valid votes cast in them. Thus, B_C depends on the geographic distribution of coalition support across districts with different electoral weights. Support concentrated in districts whose share of Chamber seats exceeds their share of the national valid vote raises the sum of district quotas relative to the national quota, while support concentrated in districts with the opposite weighting lowers it.

Thus, $d_C = A_C + B_C$. For comparisons across coalitions of different electoral sizes, the two components can also be expressed relative to the coalition's proportional quota:

$$R_C - 1 = \frac{A_C}{q_C} + \frac{B_C}{q_C}.$$

This expresses the within- and between-district contributions to the coalition's relative departure from proportionality.

In Brazil, A_C can reflect the mechanisms that operate within each district when votes are converted into seats: finite district magnitude, the electoral-quotient and highest-averages calculations, the year-specific eligibility rules for parties and candidates, and the pooling of votes through proportional electoral coalitions in 2014 and 2018 and party federations in 2022. These mechanisms determine how the coalition's actual district seats depart from its district-level proportional quotas. By contrast, B_C arises from differences in electoral weight across districts and the geographic distribution of coalition support: district apportionment, turnout, and valid-vote variation determine how many seats correspond to a given number of votes in each district, while coalition geography determines whether its support is concentrated in relatively more or less heavily weighted districts. Neither component isolates the independent effect of any one mechanism; the decomposition locates the coalition differential within or between districts rather than providing a causal decomposition of its institutional sources (Bochsler 2026).

The primary ideological universe is the set of parliamentary parties,

$$P^{+} = \{i \in P : s_i > 0\}.$$

National proportionality is evaluated over P, while potential parliamentary coalitions are constructed over P^{+}. Thus, votes for parties without seats remain part of V and of party-level proportionality accounting, but those parties cannot constitute or interrupt a primary ideological coalition. This restriction does not alter observed cabinet membership or its election-year translation.

Axelrod's connected-coalition criterion provides the baseline restriction (Axelrod 1970). Let $p_1, \ldots, p_n$ denote the existing election-year ideological order restricted to P^{+}, preserving the relative ordering of represented parties without re-estimating their positions. A coalition $C \subseteq P^{+}$ is connected when its members form a contiguous interval in this filtered order. A connected coalition is minimal winning when $s_C \geq 257$ and no proper connected subset retains a Chamber majority. For an interval, removing either endpoint must therefore leave fewer than 257 seats.

Exact connectedness ($k = 0$) provides the baseline ideological domain. As a nested sensitivity analysis, I also allow one represented party inside a coalition's ideological span to be omitted ($k = 1$). The formal definition and complete results for this one-gap domain appear in Appendix D.1.

As a sensitivity analysis, I repeat both searches in the full election-year order over P, including parties receiving votes but no seats. In that universe, such parties can be coalition members and can interrupt connectedness. Admissibility and minimality are recomputed within that order. Only the ideological universe changes: votes, seats, quotas, differentials, representation ratios, and the decomposition retain the same definitions and national vote denominator. Appendix D.2 compares the results.

3.2 Data and operationalization

Vote and seat data for the 2014, 2018, and 2022 federal-deputy elections come from the Tribunal Superior Eleitoral, accessed through the `electionsBR` package (Meireles, Silva, and Costa 2016).[3] For each party, the national vote total combines valid candidate votes and party-label votes across the twenty-seven districts, while the seat total counts deputies elected under its election-year label.[4] This attribution also applies to parties that competed through proportional electoral coalitions in 2014 and 2018 or federations in 2022. The resulting party totals connect the district electoral outcomes to the party identities observed in cabinet participation and ideological classifications.

Cabinet composition uses the pinned V6 officeholder-level chronology for the Dilma Rousseff, Michel Temer, Jair Bolsonaro, and Lula III administrations. It records service dates and contemporaneous party affiliations separately, with dated witnesses, evidence, decisions, and explicit provisional assumptions. The historical reconstruction is held fixed. I translate each dated cabinet membership into its audited election-year party identities, then treat each distinct translated set within an election as one analytical observation, including when it recurs nonconsecutively or across administrations. Membership includes all represented cabinet parties, whether or not they won Chamber seats. The linked daily chronology identifies actual appearances and cumulative duration; recurrence does not change a set's electoral quantities or its analytical weight.

The 2014 election is linked to the cabinet sequence from January 1, 2015 through December 31, 2018, and the 2018 election to the sequence from January 1, 2019 through December 31, 2022. The 2022 election is linked to the observed Lula III period, from January 1, 2023 through March

3. All 2022 results refer to the official seat vector in the frozen dataset, prior to subsequent judicial retotalizations.

4. For 2014, the vote measure is `QT_VOTOS_NOMINAIS + QT_VOTOS_LEGENDA`. For 2018 and 2022, it is `QT_VOTOS_NOMINAIS_VALIDOS + QT_TOTAL_VOTOS_LEG_VALIDOS`.

19, 2026. Renames use the audited election-year labels, while fusions map to their constituent election-year parties. Votes and seats sum the member-party totals once for each set. This yields 12 cabinet party sets linked to 2014, 18 to 2018, and 5 to 2022. Appendix B provides short labels and membership; the replication CSVs retain every analytical and historical period and actual date interval.[5]

The primary chronology covers 4,096 days: 3,996 evidence-established and 100 provisional. Nine officeholders retain historically UNKNOWN affiliations; the primary assumptions add no party on their unresolved dates. The analytical sample includes sets observed only provisionally, and retains the evidence status of every linked occurrence. Recorded concrete alternatives are evaluated separately under the same set identity rule; none changes the 260 primary inversion days. This does not bound unrecorded affiliations of unresolved people. Distinct-set frequencies are unweighted descriptions, not independent observations or probabilities of government formation. Temporal summaries use the union of actual covered dates, leaving gaps between recurrences intact.

The ideological orders for 2014 and 2018 use the party classifications of Bolognesi, Ribeiro, and Codato (2023). For 2022, I use the later 2018–2022 classification, drawing on the 2025 data release and associated article (Bolognesi et al. 2025; Bolognesi et al. 2026). Each classification supplies the election-year party order used to construct the coalition domains defined above.

4 Results

Before turning to coalitions, I first examine the party-level seat differentials d_i and representation ratios R_i produced in each election, which describe the absolute and relative departures from national proportionality that are subsequently combined at the coalition level.

In the 2014 election PMDB received 8.13 seats above its national proportional quota, the largest positive differential that year. PSD and PTB received another 4.55 and 4.37 seats above their quotas, respectively. These gains coexisted with deficits of 4.43 seats for PSDB and 4.20 for PSOL.

In 2018, PP, PR, and MDB received positive differentials of 8.76, 5.72, and 5.60 seats, while PSL and NOVO fell 7.70 and 6.35 seats below their national quotas. PSL is especially instructive: it received the largest national vote total but also the largest negative seat differential. This is not paradoxical in a multidistrict system. A large national vote total can be assembled disproportionately in districts where votes carry less weight in the national seat distribution, and party size magnifies the absolute consequence of even moderate underrepresentation. Electoral size therefore helps determine the magnitude of a party's contribution, while the direction of that contribution depends on how its votes are translated across districts.

In the 2022 election positive differentials are even more concentrated among largest parties: PL received 13.78 seats above its national proportional quota and UNIÃO another 11.10, while PT and PP gained 6.94 and 6.24 seats, respectively. These large absolute surpluses reflect both electoral size and overrepresentation. Since $d_i = q_i(R_i - 1)$, a party with a large proportional quota can contribute many seats to the positive or negative side of the national differential even when its relative departure from proportionality is moderate. In 2022, the largest parties happened predominantly to lie on the positive side of this relationship, placing substantial seat surpluses inside many of the coalition combinations available in the new Chamber.[6]

5. The replication materials are available at https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md.

6. The complete party-level vote shares, quotas, differentials, and representation ratios can be found in the replication repository.

Figure 1 shows that electoral size structures the scale of representational departures without fixing their direction. Relative representation is considerably more dispersed among small parties: at low vote shares, parties range from receiving no seats to substantial overrepresentation, while larger parties generally lie closer to $R_i = 1$. Yet the side of proportionality on which larger parties fall changes across elections. In 2014, PT and PSDB were underrepresented while PMDB, of comparable size, was overrepresented; in 2018, the largest party, PSL, was also underrepresented. The 2022 profile differs sharply: every party receiving more than 5 percent of the national vote was overrepresented, while those between 3 and 5 percent were underrepresented. The national profiles therefore show no consistent tendency for larger or smaller parties to be overrepresented. Party size instead conditions how a given relative departure translates into an absolute seat differential.

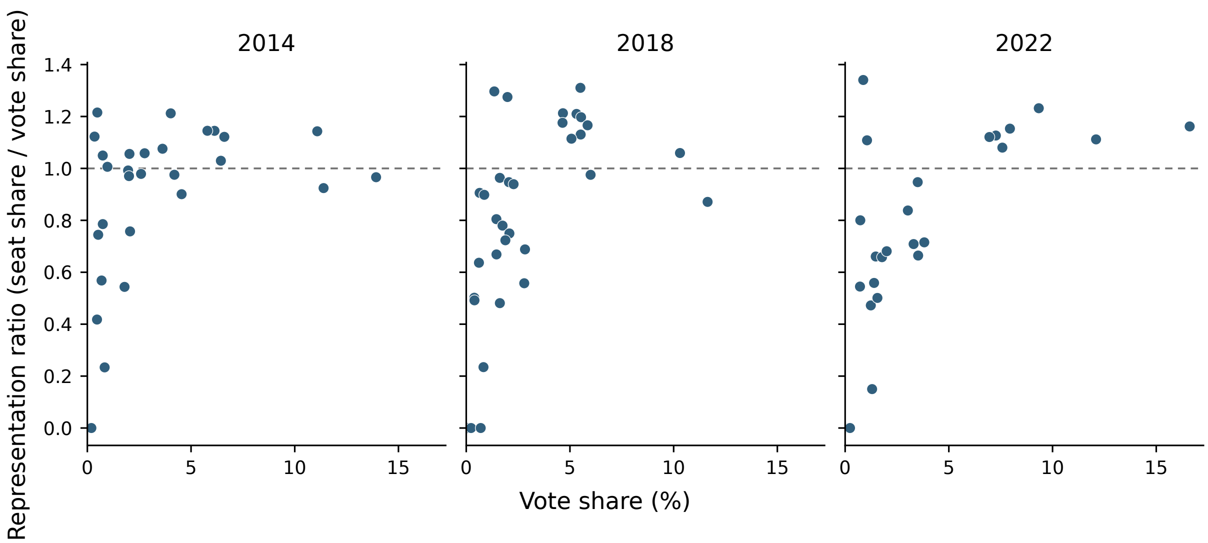

Figure 1: Party-level representation profiles, 2014, 2018, and 2022

Notes: Each point is a party. $R_i = 1$ denotes exact proportionality; parties receiving votes but no seats appear at zero.

The party profiles also clarify why representation ratios alone do not determine which parties contribute most to a coalition's seat surplus. Coalition representation is not determined by the number of overrepresented parties, nor by the largest representation ratio among its members. Each party enters in proportion to its electoral size: $R_C = \sum_{i \in C} \frac{q_i}{q_C} R_i$, and $d_C = \sum_{i \in C} d_i$. Thus, a strongly overrepresented small party may contribute relatively few seats, while a more moderately overrepresented large party may account for a substantial share of the coalition's surplus. In 2022, for example, PV had the higher representation ratio, $R_i = 1.341$, but contributed only 1.52 seats above quota; PL's lower ratio of 1.162 translated into a 13.78-seat surplus. Whether a coalition reaches a majority therefore depends on the quota-weighted combination of these party-level advantages and losses, and on whether their resulting differential is large enough to bridge the remaining distance to 257 seats.

4.1 Cabinet coalitional inversions

Two of the 35 distinct cabinet party sets combine a Chamber majority with less than half of the national federal-deputy vote. They occur under Dilma Rousseff and Lula III, following the 2014 and 2022 elections, respectively. No set observed under Michel Temer or Jair Bolsonaro inverts in the primary chronology. The two inversions differ in their distance from a vote majority, the size of their seat advantage, and their observed duration. Table 1 reports these sets and the within-district and between-district components of their differentials.

Table 1: Inverted cabinet party sets and their district components

Election	Set	Days	Vote %	Seats	d_C	A_C	B_C	Parties
2014	14-05	5	46.54	259	20.271	15.742	4.529	PCdoB, PDT, PMDB, PR, PSD, PT, PTB
2022	22-01	255	48.89	263	12.204	13.196	-0.992	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO

Days sum actual appearances of each set; all inversion days are evidence-established.

Notes: Votes, seats, and party labels refer to the corresponding election. A_C and B_C are the within-district and between-district components.

The first inversion emerges during the contraction of Dilma Rousseff's cabinet in April 2016. PRB ceases to be represented on March 18, and PP's departure on April 14 leaves a party set with 259 seats and 46.54 percent of the 2014 vote. Exact national proportionality would assign those parties 238.73 seats, leaving them 18.27 seats short of a Chamber majority. Their 20.27-seat differential, corresponding to $R_C = 1.085$, is therefore sufficient to preserve the majority. Set 14-05 is observed for five days, from April 14 through April 18. PSD's departure on April 19 reduces the party set to 223 seats and ends the inversion.

The Temer sequence does not produce an inversion. The cabinet party set observed from April 2 through April 5, 2018 comes close, with 268 seats and 50.03 percent of the 2014 vote, but it still has a vote majority. From April 6, the remaining parties have 247 seats and 45.83 percent of the vote. The contraction thus moves from a majority in both measures to a minority in both, rather than leaving a vote-minority seat majority. A positive seat differential is not, by itself, evidence of an inversion.

The cabinet sequence linked to the 2018 election also contains no inversion. Set 18-11 is observed from July 28, 2021 through March 23, 2022; its parties receive 39.79 percent of the vote and hold 222 seats. Their proportional quota is 204.14 seats, and their 17.86-seat differential corresponds to $R_C = 1.087$. They are therefore overrepresented but remain well below the Chamber majority threshold. The distinction is between receiving more seats than a coalition's vote share implies and receiving enough seats to form a majority. Here the differential falls far short of the 52.86 seats separating the coalition's proportional quota from 257 seats.

The initial Lula III cabinet provides the second inversion. Its members hold 263 seats with 48.89 percent of the 2022 vote. Their proportional quota is 250.80 seats, so a 12.20-seat differential, corresponding to $R_C = 1.049$, more than covers the 6.20 seats separating the coalition from a majority. It therefore has a larger seat majority than the Dilma coalition despite a smaller differential, because its proportional quota lies closer to 257 seats. Set 22-01 is observed for 255 days, from January 1 through September 12, 2023. The entry of PP and Republicanos on September 13 expands the party set to 350 seats and 63.79 percent of the vote. Whereas PSD's departure ends the Dilma inversion by removing the seat majority, this expansion ends the Lula inversion by supplying a vote majority as well. Within each election cycle, the underlying vote–seat distribution remains fixed; cabinet membership changes the party set across which the quotas and differentials are aggregated.

The within-district component is positive in both cabinet inversions, but the pattern extends to all 35 distinct cabinet party sets: $A_C > 0$ in every set, whereas $B_C > 0$ in thirteen. Cabinet composition helps explain the difference. Parties represented at least once in the primary cabinet chronology are larger on average than parties never represented, with mean vote shares of 5.72 versus 0.83 percent in 2014, 5.41 versus 1.84 percent in 2018, and 5.57 versus 1.66 percent in 2022. Larger parties are also usually on the positive side of the within-district component. Among parties receiving at least 5 percent of the national vote, $A_i > 0$ for six of seven in 2014, eight of nine in 2018, and all seven in 2022.[7] In 34 of the 35 distinct election-year cabinet party sets, the positive A_i contributions of members meeting this benchmark exceed all negative A_i contributions combined. In the remaining set, DEM, PSC, PSD, and PSL, positive contributions from smaller parties are also needed to keep A_C positive.[8] Cabinet coalitions therefore repeatedly accumulate a positive within-district component, but not exclusively through their largest members. No equivalent sign pattern appears for B_C, which may reinforce or offset that advantage depending on the distribution of coalition support across districts.

The two inverted cabinet party sets differ in how the positive A_C interacts with between-district weighting. For 14-05, $A_C = 15.74$ raises the coalition's quota of 238.73 to $q_C + A_C = 254.47$ seats, still short of a Chamber majority. Adding $B_C = 4.53$ brings the observed total to 259 seats. For 22-01, by contrast, $A_C = 13.20$ already exceeds the 6.20 seats separating the quota from a majority: $q_C + A_C = 263.99$. The negative between-district component, $B_C = -0.99$, reduces the total to 263 seats without removing the majority. Thus, within-district gains characterize both inversions, while between-district weighting reinforces the Dilma coalition and offsets the Lula coalition. Only in the former is the positive between-district component also needed to bridge the proportional distance to 257 seats.

The distinct cabinet party sets show that overrepresentation is much more common than inversion. Figure 2 shows that 33 of the 35 sets have $R_C > 1$; the two underrepresented sets are linked to the 2018 election. Set 18-03 recurs in periods 2020.2 and 2020.4, while 18-18 is observed only on the two provisional days of period 2022.7. Most overrepresented cabinet party sets do not reverse majority status: some remain below both 50 percent of the vote and 257 seats, while others exceed both. The two inversions are the sets for which these thresholds diverge. Each set appears once in the figure, ordered by its first appearance; the figure does not show duration or recurrence.

Measured over time, inversions occupy 5 of the 1,461 covered days following the 2014 election, none of the 1,461 following the 2018 election, and 255 of the 1,174 observed days following the 2022 election. Inverted cabinet configurations were therefore in office for 260 of the 4,096 covered days in the primary chronology. The two episodes range from a brief cabinet contraction in 2016 to the first several months of Lula III. Their endings illustrate two different transitions: a coalition can lose its seat majority or acquire a vote majority. The same fixed electoral distribution can therefore sustain different majority relations as parties are recombined in cabinet.

7. The parties receiving at least 5 percent with positive A_i are PSDB, PMDB, PP, PSB, PSD, and PR in 2014; PT, PSDB, PSD, MDB, PSB, PP, PR, and PRB in 2018; and PL, PT, UNIÃO, PP, PSD, MDB, and Republicanos in 2022. The exceptions are PT in 2014 and PSL in 2018.

8. Set 18-08, observed from December 9, 2020 through February 11, 2021.

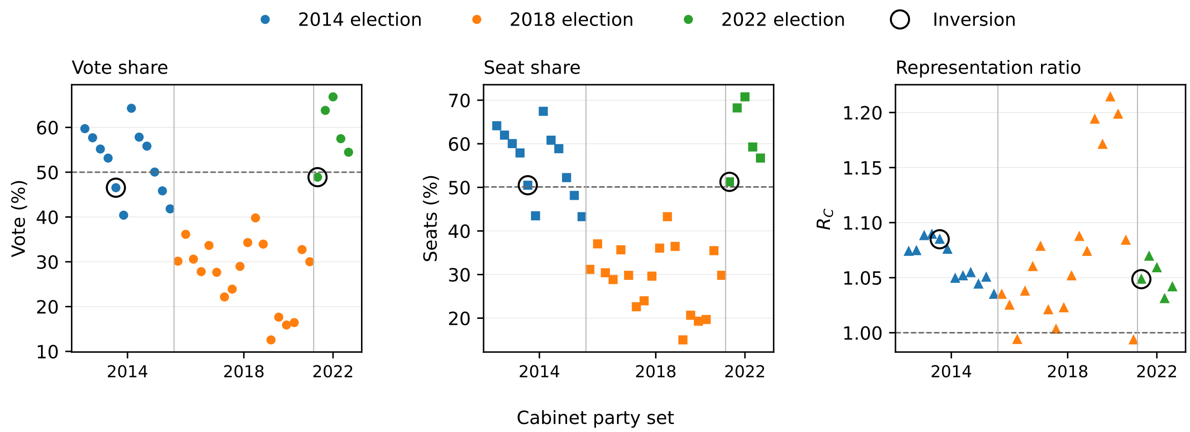

Figure 2: Cabinet party-set vote shares, seat shares, and representation ratios

4.2 Ideologically constrained potential coalitions

There are minimal connected winning inversions in all three elections: four of eight minimal connected winning coalitions invert in 2014, one of eight in 2018, and two of four in 2022. These coalitions are constructed over 28, 30, and 23 seat-winning parties, respectively. The counts establish that inversions occur in each election; the figures below show where those cases lie in the ideological ordering and how they relate to the wider set of connected majorities. Figure 3 identifies the minimal inversion intervals themselves, while Figure 4 places them within the full exact-connected interval space.

PT provides a useful substantive reference point within these ideological intervals. It is the principal electoral and organizational pole of the Brazilian left and one of the largest parties in each of the legislatures considered. It also held the presidency at the beginning of the 2015–2018 mandate under Dilma Rousseff and returned to the presidency with Lula following the 2022 election. Figure 3 marks PT in each election-year ordering as a common substantive reference point. Whether a connected majority includes PT therefore indicates whether the coalition extends through the principal left-wing pole of the party system or can instead be assembled entirely to its right. Because the ideological classifications are used as orderings rather than metric scales, references below to broader or narrower intervals concern the number of adjacent party positions spanned, not cardinal ideological distance.

The four 2014 connected inversions contain twelve to fifteen parties. PSB–PTN begins on the center-left; the other three begin at parties classified as center-right, and all extend into the right. None includes PT, despite PT holding the presidency under Dilma Rousseff and anchoring the left side of the party system. The connected inversion region instead begins to its right: every case combines PMDB and PSD with PSDB. The strongest inversion, measured by the lowest vote share, is the thirteen-party PTB–PR interval: 47.45 percent of all valid votes yields exactly 257 seats, a quota of 243.40 seats, and a differential of 13.60 seats.

The 2018 configuration differs. PT–PSDB is the only exact-connected minimal inversion, so the inversion interval now extends from the principal party of the left through PSDB rather than excluding PT altogether. Its fourteen parties receive 49.95 percent of all valid votes and hold 267 seats. Figure 3 makes an important distinction visible: this is a broad interval in terms of the number of party positions it spans, but its vote share lies only 0.05 percentage points below one half. The proportional quota is 256.25 seats and the differential is 10.75 seats. The other seven minimal connected majorities have vote majorities, including both overrepresented and underrepresented combinations. This election therefore supplies a potential inversion very close to the vote-majority threshold under exact connectedness, even though no cabinet party set linked to the same election inverts in the primary chronology.

In 2022, the non-inverted minimal majorities PT–PP and PSB–PSC begin on the left and center-left and contain fifteen and sixteen parties, respectively. They retain seat majorities despite negative differentials of 9.36 and 15.10 seats. The inversions occur in the ten-party MDB–UNIÃO interval, spanning the center-right through the far right, and the seven-party PP–PL interval, confined to the right and far right.

MDB–UNIÃO receives 49.22 percent of the vote, placing its proportional quota at 252.49 seats, 4.51 below the majority threshold. Its 12.51-seat differential yields 265 seats. PP–PL receives 45.26 percent of the vote and has a quota of 232.19 seats, leaving a 24.81-seat shortfall from 257. Its larger differential of 25.81 seats yields a smaller majority of 258 seats. PL and UNIÃO contribute 13.78 and 11.10 seats above quota to this latter coalition, which also includes underrepresented members. The contrast shows why the size of a seat majority cannot be read directly from the extent of overrepresentation: a coalition with less electoral support requires a larger advantage to reach the same threshold. Full $k = 0$ minimal-winning compositions appear in Appendix C.

Figure 3 makes the membership and endpoint structure of the seven cases visible. The four 2014 inversions are not isolated combinations but a family of heavily overlapping center-to-right intervals, all excluding PT and sharing much of the same party core. The 2018 case is qualitatively different: the sole minimal inversion extends from PT through PSDB and spans fourteen adjacent parties, even though its vote share falls only marginally below one half. In 2022, both minimal inversions again lie entirely to PT's right, but they differ sharply in span and electoral support: MDB–UNIÃO covers a broader center-right-to-far-right interval, whereas PP–PL reaches a seat majority with only seven adjacent parties and 45.26 percent of the vote. The location of PT as a reference point helps distinguish the different positions of the coalition inversion region within the ideological order. The figure also separates two features that need not coincide: the breadth of an interval in the party ordering and the size of its vote deficit. Figure 4 then shows whether these minimal cases are isolated or sit on the boundary of larger regions of connected majorities.

Figure 4 complements Figure 3 by placing the minimal inversions within the full set of ideologically-connected intervals. Blue cells identify connected coalitions with both a Chamber majority and a vote majority, red cells identify minimal connected winning inversions, and orange cells identify larger connected coalitions that remain inverted. Whereas Figure 3 identifies the membership and endpoints of the minimal cases, the heatmaps show how those cases sit within the surrounding space of connected coalitions.

The structure differs substantially across elections. In 2014, several minimal inversions lie in overlapping center-to-right spans, and some are accompanied by orange cells. The inversion is therefore not confined to the minimal winning coalitions themselves: some larger connected intervals also retain a seat majority while remaining below 50 percent of the national vote. In 2018, the blue region of vote-and-seat majorities contracts and its boundary shifts toward intervals extending farther to the right of the party ordering. At the same time, PT–PSDB is the only inversion and there are no orange cells. This is a distinct feature of the 2018 configuration. The PT–PSDB interval has 49.95 percent of the vote and 267 seats, but any larger exact-connected interval containing it also has a vote majority. The single inversion is therefore isolated within the surrounding connected coalition space rather than forming part of a larger set of inverted intervals.

The 2022 pattern differs again. The blue region is smaller, reflecting a lower proportion of connected coalitions with both vote and seat majorities, while the inversion cells are concentrated

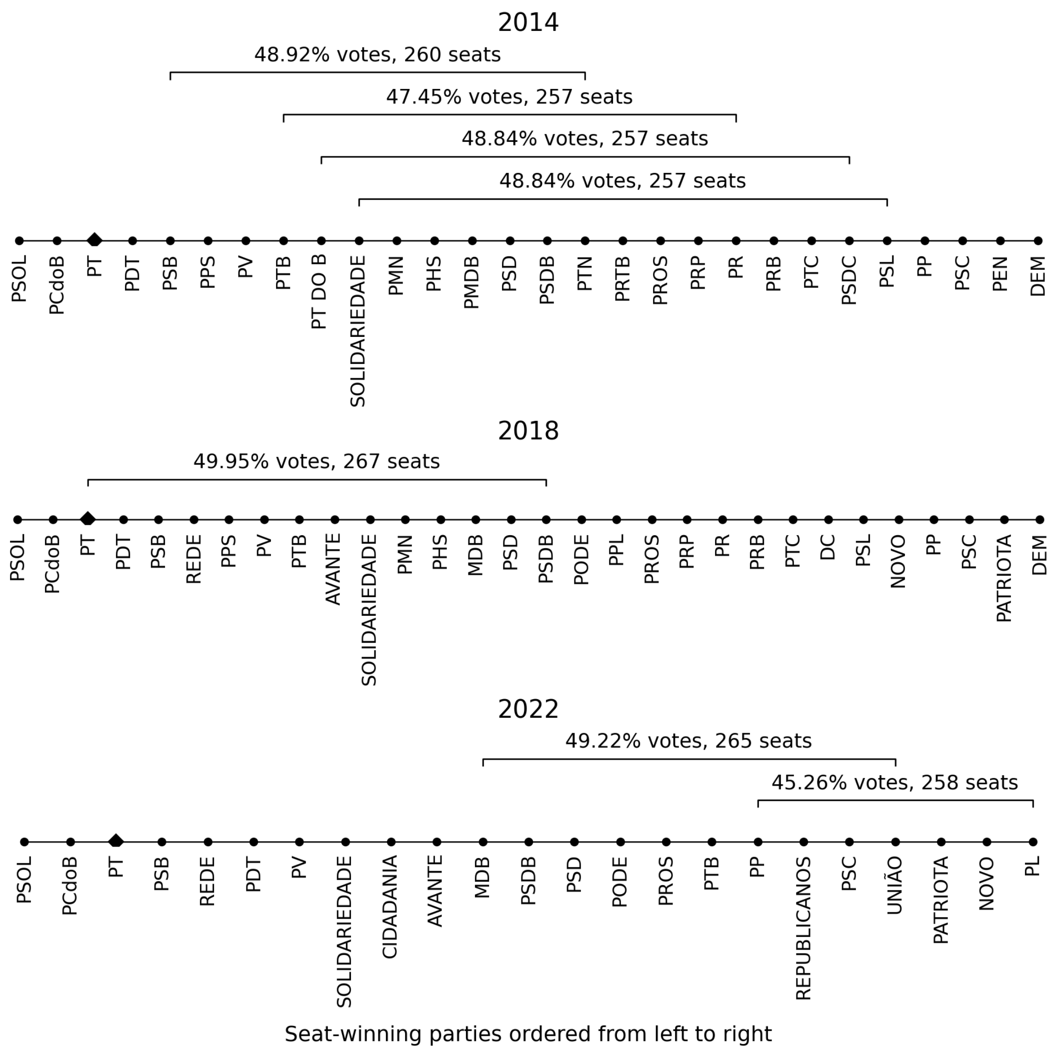

Figure 3: Minimal connected winning inversions by election year. Brackets identify coalition endpoints and report vote shares and seats; the diamond marks PT as a common reference point.

on the right. Unlike 2018, orange cells reappear: some connected expansions of the minimal right-side inversions continue to hold a Chamber majority without reaching 50 percent of the national vote. Connected majorities extending through PT, by contrast, require broader intervals and have both vote and seat majorities. The 2022 result is therefore not merely the existence of two minimal inversions, but an asymmetry in where connected vote-and-seat majorities and connected vote-minority seat majorities occur within the party ordering.

Figure 3 and Figure 4 thus reveal different aspects of the same result: the first identifies the coalitions at the minimal winning boundary, while the second shows whether inversion remains present as the connected interval is expanded. The right-side concentration in 2022 is consistent with the party-level results above, where large right and center-right parties, particularly PL and UNIÃO, carry substantial positive seat differentials that can reinforce one another when adjacent parties are combined.

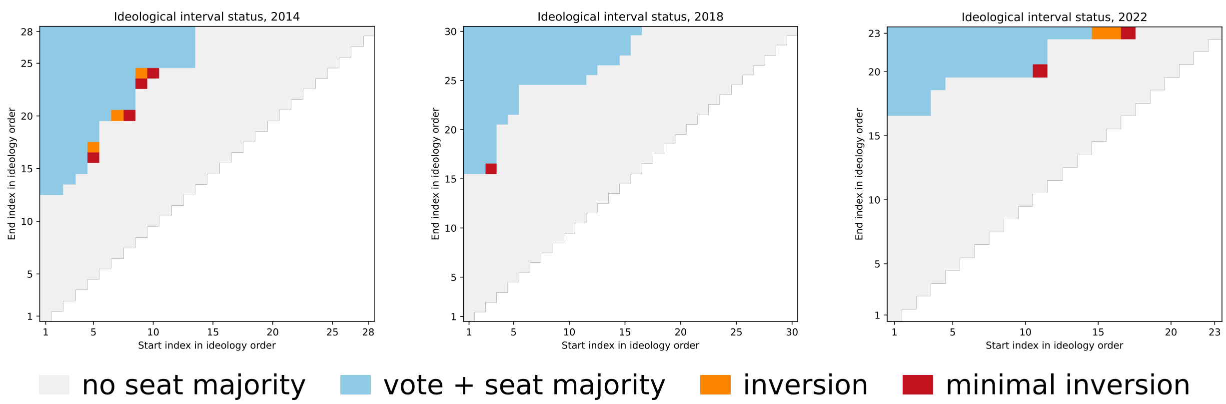

Figure 4: Exact-connected ($k = 0$) ideological interval status by election

Note: Start and end indices refer to the election-year ideological order of seat-winning parties.

Allowing one represented party inside a coalition's ideological span to be omitted ($k = 1$) does not overturn the main pattern: minimal inversions remain present in all three elections. The strongest 2014 and 2022 cases retain the same member sets and vote–seat outcomes. In 2018, the relaxation produces a stronger inversion than exact connectedness: PCdoB–PODE omitting PSDB, with 47.59 percent of the vote and 257 seats. Appendix D.1 reports the formal definition and complete one-gap robustness results. A separate sensitivity to the ideological party universe preserves the strongest 2014 and 2022 regions but removes the 2018 exact-connected inversion when parties without seats are restored to the ideological order; Appendix D.2 reports the comparison.

4.3 Comparing decomposition patterns across coalition domains

The inversions identified in the two coalition domains share a positive within-district component, but differ in how between-district weighting contributes to their seat majorities. Table 2 reports the decomposition for the minimal connected inversions, complementing the cabinet results in Table 1. Reading the tables together separates two questions: which component accounts for most of a coalition's seat differential, and whether that component is sufficient to cover the distance from its proportional quota to 257 seats. A component can account for most of the differential without being sufficient to produce the seat majority.

Table 2: Minimal connected winning inversions and seat-differential decomposition

Election	Start	End	Parties	Vote %	Seats	d_C	A_C	B_C
2014	PSB	PTN	12	48.92	260	9.032	10.509	-1.477
2014	PTB	PR	13	47.45	257	13.601	12.507	1.094
2014	PT DO B	PSDC	15	48.84	257	6.442	7.153	-0.711
2014	SOLIDARIEDADE	PSL	15	48.84	257	6.462	7.473	-1.011
2018	PT	PSDB	14	49.95	267	10.751	6.568	4.183
2022	MDB	UNIÃO	10	49.22	265	12.507	2.953	9.554
2022	PP	PL	7	45.26	258	25.813	23.321	2.492

The A_i contributions of individual coalition members help explain the persistent positive within-district component of cabinet coalitions.[9] In the 2016 cabinet inversion, PR and PSD contribute 5.04 and 4.65 seats to A_C. In the initial 2023 cabinet, UNIÃO and PT contribute 9.36 and 9.07 seats. Cabinet membership repeatedly brings substantial positive A_i contributions into the same coalition. As the party-size comparison in Section 4.1 shows, large members usually supply enough positive contributions to offset the negative ones, but smaller members also matter. The resulting A_C is positive in all 35 distinct cabinet party sets, including the two inversions.

Cabinet coalitions show no comparable tendency in the between-district component: B_C is positive in thirteen of the 35 distinct cabinet party sets. The two components can also pull in opposite directions for the same party. In the initial 2023 cabinet, for example, PT contributes 9.07 seats to A_C but -2.13 seats to B_C. Among the 2014 inversions, between-district weighting reinforces the within-district component in the Dilma cabinet and the PTB–PR interval, but offsets it in the other three minimal connected inversions. For the initial Lula cabinet, it also offsets the within-district gain.

The 2018 election permits a comparison between an observed cabinet minority and a potential connected inversion. Cabinet party set 18-11 combines $A_C = 21.41$ with $B_C = -3.55$, but its 222 seats do not constitute a majority. The PT–PSDB connected inversion instead combines positive contributions from both components, $A_C = 6.57$ and $B_C = 4.18$. Its proportional quota lies only 0.75 seats below the majority threshold, so either component individually exceeds that shortfall. The cabinet's larger within-district gain therefore does not produce the same majority status: its quota lies much farther below 257 seats. The same electoral result thus permits a cabinet minority with a larger within-district gain and a connected seat majority with a much smaller proportional shortfall.

MDB–UNIÃO therefore provides the exceptional case within the minimal connected winning inversions: it is the only case in which between-district weighting accounts for more of the coalition's seat surplus than within-district allocation. Its 12.51-seat differential consists of $A_C = 2.95$ and $B_C = 9.55$. The relatively small A_C results from substantial cancellation among its members: positive A_i contributions from UNIÃO and MDB are largely offset by negative contributions from PTB and PODEMOS (Appendix A.1). Its positive B_C, by contrast, indicates that the coalition's support is distributed relatively toward districts with greater electoral weight. Since its proportional quota of 252.49 seats lies only 4.51 seats below a majority, the between-district component alone is larger than that gap. MDB–UNIÃO is therefore the only inversion whose seat differential is primarily associated with favorable between-district weighting rather than within-district allocation.

PP–PL shows why the component that accounts for most of a coalition's seat differential need not, by itself, span the distance from its proportional quota to 257 seats. Its $A_C = 23.32$

9. The largest member-party A_i and B_i contributions for the focal coalitions are reported in Appendix A.1.

accounts for most of its 25.81-seat differential, while $B_C = 2.49$ is comparatively small. Yet the coalition's proportional quota is only 232.19 seats, leaving it 24.81 seats short of a majority. The within-district component therefore falls short of that distance on its own; adding the positive between-district component brings the observed total to 258 seats. The Dilma cabinet inversion and the 2014 PTB–PR interval have the same structure: A_C is the larger component, but both positive components enter the differential that carries the coalition across 257.

The sensitivity analysis for ideological coalitions separates two features that coincide in most of the cabinet and minimal-connected winning cases. Positive within-district contributions remain pervasive: $A_C > 0$ in ninety-seven of the one hundred minimal inversions under the one-gap relaxation. But within-district allocation is not always the larger component. $A_C > B_C$ in seventy-seven of those one hundred cases, with substantial variation across elections: the relationship holds in forty of forty-six inversions in 2014 and thirty-two of forty-two in 2018, but only five of twelve in 2022 (Appendix D.1). Thus, the predominance of within-district gains remains strong overall, while the 2022 inversions reveal a different coalitional pattern in which favorable between-district weighting often provides the larger contribution to the seat surplus.

Restoring parties without seats to the ideological order reinforces the need to distinguish these two patterns. MDB–UNIÃO then combines $A_C = -1.29$ with $B_C = 9.88$ and remains inverted (Appendix D.2). The decomposition therefore shows both a persistent tendency for inverted coalitions to accumulate within-district gains and meaningful variation in whether those gains or between-district weighting account for the larger share of the resulting seat differential.

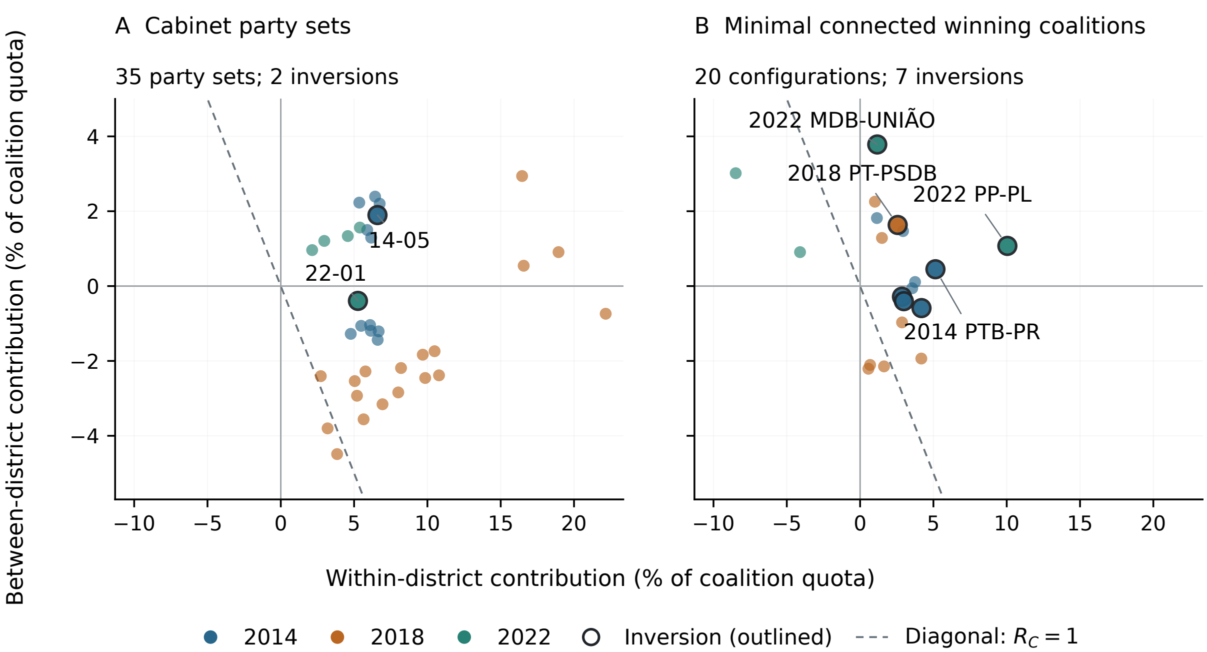

Figure 5: Within-district and between-district contributions across coalition domains

Figure 5 extends the comparison beyond the inverted coalitions. Panel A includes all 35 distinct election-year cabinet party sets, including those observed under the provisional affiliation assumptions described in Section 3.2. Panel B includes all twenty minimal connected winning coalitions in the primary ideological domain. The horizontal and vertical coordinates are $100A_C/q_C$ and $100B_C/q_C$, respectively. Their sum is $100(R_C - 1)$, so coalitions with the same representation ratio lie along parallel lines with slope -1. The dashed diagonal identifies $R_C = 1$. The figure therefore shows something that R_C alone cannot: coalitions with similar overall representation can reach that result through different balances of within-district and between-district contributions. The diagonal is not an inversion boundary. Inversion additionally requires less than half of the national vote and at least 257 Chamber seats.

Panel A shows that a positive within-district component characterizes all distinct cabinet party sets, including the inverted cases. Every one of its 35 sets lies to the right of $A_C = 0$, including the two underrepresented sets linked to the 2018 election. For those sets, negative between-district contributions more than offset the within-district gain. The two inverted cabinets are not those with the largest A_C/q_C: non-inverted sets also lie farther to the right. A large within-district contribution relative to the coalition's quota is therefore not sufficient for inversion. The majority condition also depends on the size of that quota and on B_C.

Panel B shows a different pattern. Minimal connected winning coalitions appear on both sides of $A_C = 0$, so the uniformly positive within-district component found among cabinets does not extend to the ideological domain as a whole. The 2022 cases provide the clearest comparison. All four minimal connected majorities have $B_C > 0$, but the two vote-majority coalitions, PT–PP and PSB–PSC, have negative A_C large enough to leave them underrepresented overall. MDB–UNIÃO and PP–PL instead have positive contributions from both components and reach seat majorities with vote minorities. Favorable between-district weighting is therefore shared by all four 2022 minimal majorities; what separates the inverted from the non-inverted cases is how it combines with the within-district component and with each coalition's distance from the majority threshold.

MDB–UNIÃO is especially distinctive. Among all baseline cabinet and minimal-connected inversions, it is the only case in which the between-district component exceeds the within-district component: $B_C = 9.55$ seats compared with $A_C = 2.95$. Its proportional quota is 252.49 seats, only 4.51 below the Chamber majority. The within-district contribution does not cover that distance, whereas the between-district contribution does. PP–PL presents the opposite structure. Its much larger within-district component, $A_C = 23.32$, supplies most of the 25.81-seat differential required to raise a coalition with only 45.26 percent of the vote to 258 seats. The two 2022 inversions therefore reach the same majority status through sharply different balances between the components.

The initial Lula cabinet provides a final cross-domain comparison. Its representation ratio, approximately 1.049, is almost identical to the 1.050 ratio of MDB–UNIÃO, yet the two coalitions occupy very different positions in Figure 5. For the cabinet, A_C/q_C and B_C/q_C are approximately 5.26 and -0.40 percent; for MDB–UNIÃO they are 1.17 and 3.78 percent. The same 2022 election therefore produces nearly identical overall coalition representation through very different balances between the two components. Neither the election year nor R_C by itself determines how a coalition's representational advantage is composed.

5 Discussion

The analysis compares coalition inversions across two politically defined domains in a proportional system institutionally different from the principal cases in the existing literature. The first domain follows the parties actually represented in cabinet as composition changes during a mandate. The second enumerates ideologically constrained potential coalitions using exact connectedness as the baseline. The same party vote–seat distribution can therefore be evaluated under realized and structured potential recombinations.

The distinction matters because coalition inversion is configuration-dependent. No observed cabinet party set linked to 2018 inverts, while the PT–PSDB connected interval does. The one-gap robustness check preserves the strongest 2014 and 2022 compositions and outcomes, while 2018 is more sensitive to admissibility: the relaxation yields a stronger inversion, and the all-party universe removes the exact-connected case (Appendix D). Observed cabinets and ideologically constrained potential coalitions therefore reveal different majority configurations within the same electoral distribution.

The party representation profiles also show why coalition inversion should not be reduced to one stable party-size bias. In 2014, overrepresentation and underrepresentation are distributed across parties of different sizes. In the 2022 data, larger parties are more consistently overrepresented, but coalition outcomes still depend on which positive and negative party differentials are combined. The political consequence of a fixed electoral result is realized through the party sets that organize legislative and governing majorities.

At the coalition level, R_C provides the bridge between party representation and majority status. It is a quota-weighted aggregation of the member-party ratios, not a simple count of overrepresented parties. The cabinet comparison shows that relative advantage is common: 33 of 35 party sets have $R_C > 1$, but only two invert. Majority-changing advantage depends jointly on the size of $R_C - 1$ and on q_C, because the latter fixes how many surplus seats are needed to reach 257. The contrast between the MDB–UNIÃO interval, whose quota is closer to a majority, and the more strongly overrepresented PP–PL interval makes this distinction especially clear.

The decomposition of the exact-connected domain adds a second comparative result. The within-district component is positive in both inverted cabinet party sets and all seven minimal connected inversions. Between-district weighting reinforces it in the PTB–PR, PT–PSDB, MDB–UNIÃO, and PP–PL ideological cases and in cabinet party set 14-05; it offsets it in the other three 2014 ideological inversions and cabinet party set 22-01. The balance between components varies even when every case satisfies the same inversion criterion.

These comparisons locate national differentials in the decomposition without assigning causal responsibility. B_C is not pure malapportionment; it combines district seat weights, turnout and valid-vote differences, and coalition vote geography. Likewise, party contributions in 2014 and 2018 are ex post attributions in elections where seats were often allocated through joint electoral lists. A large positive or negative party or district entry does not establish that the unit caused an inversion.

Brazil's multiple sources of disproportionality do not undermine the aggregation argument, but they delimit its interpretation. The analysis does not show that district magnitude, malapportionment, turnout, electoral coalitions, federations, thresholds, or rounding individually produced an inversion. It shows that the realized party differentials generated by this institutional environment can accumulate across particular political sets and cross a discontinuous majority threshold. In a pure-PR benchmark the whole-number problem is sufficient for this possibility; in Brazil it is one potential source among several.

Institutional complexity therefore weakens a monocausal claim but strengthens the case for studying the coalition transformation itself. Party differentials that appear modest or politically inconsequential in isolation may reinforce one another inside one coalition, cancel inside another, and leave a third just below 257 seats. The relevant contribution is not a claim that Brazil reveals a new source of disproportionality. It is the demonstration that ordinary differentials from several sources acquire a distinct institutional consequence when parties are recombined into realized or constrained potential majorities.

This limited claim nevertheless places the Brazilian cases within the broader literature on paradoxes and tradeoffs under proportional representation. Party-level proportionality does not settle the majority status of every post-electoral party set. Coalition inversions identify the narrower circumstance in which the ordinary vote–seat translation, once aggregated across a politically relevant set, yields a legislative majority without a corresponding national vote majority (Van Deemen 1993; Kurrild-Klitgaard 2008; Kaminski 2018; Flis, Grofman, and Kaminski 2025).

Two further qualifications remain. Both the exact-connected baseline and the one-gap robustness check condition on a fixed estimated party order. They therefore do not incorporate uncertainty in the ideology estimates themselves; alternative placements may change endpoints, spans, or marginal cases. In addition, the vote share of a cabinet or potential coalition is the sum of votes cast for parties later grouped into that set, not a direct vote for the coalition. The criterion measures coalitional representation; it does not measure voter endorsement, legislative discipline, roll-call support, or portfolio bargaining. The analysis identifies a representational condition, not its public reception or normative legitimacy.

6 Conclusion

This article extends the literature on coalition inversions by following cabinet composition across legislative terms, comparing those realized party sets with ideologically constrained potential coalitions, and examining both in a multidistrict proportional system without national compensation. Party-level deviations can aggregate into politically consequential coalition differentials, but whether they change majority status depends on coalition configuration.

The empirical comparison is not uniform across elections or domains. Two of the 35 distinct cabinet party sets invert: the Dilma set observed for five days following the 2014 election and the initial Lula set observed for 255 days following the 2022 election. No primary cabinet set linked to 2018 inverts. Inversions occur among minimal connected winning coalitions in all three elections. PP–PL provides a compact seven-party right-side inversion in 2022, while PT–PSDB has a narrow vote minority in 2018. Relaxing exact connectedness preserves the strongest 2014 and 2022 results and reveals stronger 2018 inversion possibilities, while the all-party sensitivity removes the exact-connected 2018 case.

The coalition representation ratio clarifies the threshold logic. Most observed cabinet coalitions are overrepresented, but only two combine that advantage with a vote minority and a seat majority. R_C measures relative advantage, d_C measures its absolute seat value, and q_C determines the coalition's proportional distance from 257. Coalition inversion is therefore not synonymous with overrepresentation. It is the subset of overrepresented configurations in which the accumulated seat surplus is majority-changing.

Brazil's multiple sources of disproportionality do not undermine the aggregation argument, but they delimit its interpretation. The analysis does not show that district magnitude, malapportionment, turnout, electoral coalitions, federations, thresholds, or rounding individually produced an inversion. It shows that the realized party differentials generated by this institutional environment can accumulate across particular political sets and cross a discontinuous majority threshold. In a pure-PR benchmark the whole-number problem is sufficient for this possibility; in Brazil it is one potential source among several.

Institutional complexity therefore weakens a monocausal claim but strengthens the case for studying the coalition transformation itself. Party differentials that appear modest or politically inconsequential in isolation may reinforce one another inside one coalition, cancel inside another, and leave a third just below 257 seats. The relevant contribution is not a claim that Brazil reveals a new source of disproportionality. It is the demonstration that ordinary differentials from several sources acquire a distinct institutional consequence when parties are recombined into realized or constrained potential majorities.

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Data availability. For peer review, the data and replication materials are available in an anonymized repository: https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md. For final publication, the analysis uses public election data from Brazil's Tribunal Superior Eleitoral and cabinet/party classification inputs documented in the replication repository.

Code availability. For peer review, the code necessary to reproduce the analysis is available in an anonymized repository: https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md. For final publication, the replication code is available at the public repository after acceptance.

Ethics approval. This study uses publicly available aggregate electoral, cabinet, and party-classification data. No new human participants were recruited by the author.

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A Seat-differential diagnostics

Party contributions identify which coalition members add to or offset d_C. The A_C and B_C terms partition the coalition differential into within-district allocation and between-district weighting, while district electoral weights provide a separate institutional diagnostic. These quantities are descriptive and do not identify causal effects.

A.1 Party contributions to coalition differentials

Table 3 reports the member-party composition of d_C for the focal cabinet and primary minimal parliamentary inversions. The complete machine-readable output includes all two inverted cabinet party sets and all twelve primary exact-connected $k = 0$ ideological inversions, including the seven minimal cases, together with the all-party sensitivity results.

Table 3: Party contributions to coalition differentials

Case	d_C	Gross +	Gross -	Two largest positive d_i	Two largest negative d_i
Cabinet 2014/14-05	20.271	22.773	2.502	PMDB (8.13); PSD (4.55)	PT (-2.42); PCdoB (-0.08)
Cabinet 2022/22-01	12.204	26.461	14.257	UNIÃO (11.10); PT (6.94)	PSOL (-6.06); PSB (-5.57)
Ideological 2014/PSB-PTN	9.032	19.612	10.580	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/PTB-PR	13.601	23.526	9.925	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/PT DO B-PSDC	6.442	19.369	12.927	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/SOLIDARIEDADE-PSL	6.462	19.369	12.907	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PSL (-3.26)
Ideological 2018/PT-PSDB	10.751	24.401	13.650	MDB (5.60); PSD (4.99)	PV (-4.32); REDE (-3.26)
Ideological 2022/MDB-UNIÃO	12.507	29.495	16.988	UNIÃO (11.10); PP (6.24)	PTB (-5.67); PODE (-4.94)
Ideological 2022/PP-PL	25.813	35.441	9.628	PL (13.78); UNIÃO (11.10)	NOVO (-3.35); PATRIOTA (-3.16)

Notes: Gross - is the absolute sum of negative party contributions. The three aggregate columns use a closure-preserving three-decimal display; all identities are calculated and checked exactly before rounding. d_i is an ex post accounting contribution to the coalition differential. Positive and negative party contributions sum to d_C. For 2014 and 2018, party-level contributions are ex post accounting attributions because seats were often allocated through joint electoral lists. These values do not identify party-specific causal effects of the electoral rules.
V5 primary sets include 100 flagged provisional days; the underlying UNKNOWN affiliations remain unresolved. Recorded concrete date and affiliation sensitivities are reported separately.

Table 4 reports the largest positive and negative A_i and B_i contributions among the members of the same focal coalitions.

Table 4: Largest member-party contributions to the within- and between-district components

Case	Within-district A_i		Between-district B_i	
	Positive	Negative	Positive	Negative
Observed cabinet inversions				
2014/14-05	PR (5.04) PSD (4.65)	PT (-1.22) PCdoB (-0.57)	PMDB (4.22) PDT (1.56)	PT (-1.21) PR (-0.74)
2022/22-01	UNIÃO (9.36) PT (9.07)	PSB (-5.55) PSOL (-3.40)	UNIÃO (1.75) PDT (1.02)	PSOL (-2.66) PT (-2.13)
Minimal exact-connected ideological inversions ($k = 0$)				
2014/PSB–PTN	PSD (4.65) PTB (4.08)	PT DO B (-3.61) PV (-1.40)	PMDB (4.22) PMN (0.36)	PSDB (-5.01) PV (-1.16)
2014/PTB–PR	PR (5.04) PSD (4.65)	PT DO B (-3.61) PRTB (-1.91)	PMDB (4.22) PROS (0.76)	PSDB (-5.01) PR (-0.74)
2014/PT DO B–PSDC	PR (5.04) PSD (4.65)	PT DO B (-3.61) PRTB (-1.91)	PMDB (4.22) PROS (0.76)	PSDB (-5.01) PRB (-2.17)
2014/SOLIDARIEDADE–PSL	PR (5.04) PSD (4.65)	PSL (-3.30) PRTB (-1.91)	PMDB (4.22) PROS (0.76)	PSDB (-5.01) PRB (-2.17)
2018/PT–PSDB	PT (4.88) MDB (4.34)	PV (-4.01) REDE (-3.94)	SOLIDARIEDADE (1.81) MDB (1.26)	PT (-1.75) PSDB (-1.18)
2022/MDB–UNIÃO	UNIÃO (9.36) MDB (5.12)	PTB (-5.94) PODE (-5.66)	PP (3.17) REPUBLICANOS (2.94)	PSDB (-0.54) MDB (-0.40)
2022/PP–PL	PL (19.07) UNIÃO (9.36)	PSC (-3.71) PATRIOTA (-3.03)	PP (3.17) REPUBLICANOS (2.94)	PL (-5.29) NOVO (-0.55)

Notes: Entries show the two largest positive and two most negative contributions in seats; values are rounded to two decimals. Cases are identified by election year and cabinet party-set label or interval endpoints. A_i and B_i are ex post accounting contributions: $d_i = A_i + B_i$, $A_C = \sum_{i \in C} A_i$, and $B_C = \sum_{i \in C} B_i$. For 2014 and 2018, party-level attribution is ex post because seats were often allocated through joint electoral lists. These values do not identify party-specific causal effects of electoral rules.
V5 primary sets include 100 flagged provisional days; the underlying UNKNOWN affiliations remain unresolved. Recorded concrete date and affiliation sensitivities are reported separately.

A.2 District contributions to cabinet inversions

Table 5 summarizes the district contributions to A_C in the two inverted cabinet party sets. Positive and negative sums are calculated separately across districts. Concentration is measured relative to the positive sum, $\sum_d \max(A_{Cd}, 0)$, before subtracting negative contributions.

Table 5: District contributions to the within-district component of cabinet inversions

Election	Set	Districts positive/negative	Positive sum	Negative sum	Top three share (%)	Districts for 90%
2014	14-05	20/7	20.42	-4.67	45.1	13
2022	22-01	18/9	20.84	-7.64	37.4	12

V5 primary sets include 100 flagged provisional days; the underlying UNKNOWN affiliations remain unresolved. Recorded concrete date and affiliation sensitivities are reported separately.

Notes: Sums are measured in seats. Concentration columns refer to positive district contributions only.

A.3 District electoral weights

The neutral district-weight diagnostic compares each district's seats per valid vote with the national rate:

$$w_d = \frac{S_d/V_d}{S/V} = \frac{S_d/S}{V_d/V}.$$

Figure 6 shows an interpretable but only moderate inverse relationship between district magnitude and electoral weight. São Paulo is the unique 70-seat, lowest-weight district in all three elections, whereas the eleven eight-seat districts span values below the national average to more than five times it. Magnitude alone therefore does not define a homogeneous weighting category, and the plot is not coalition-specific.

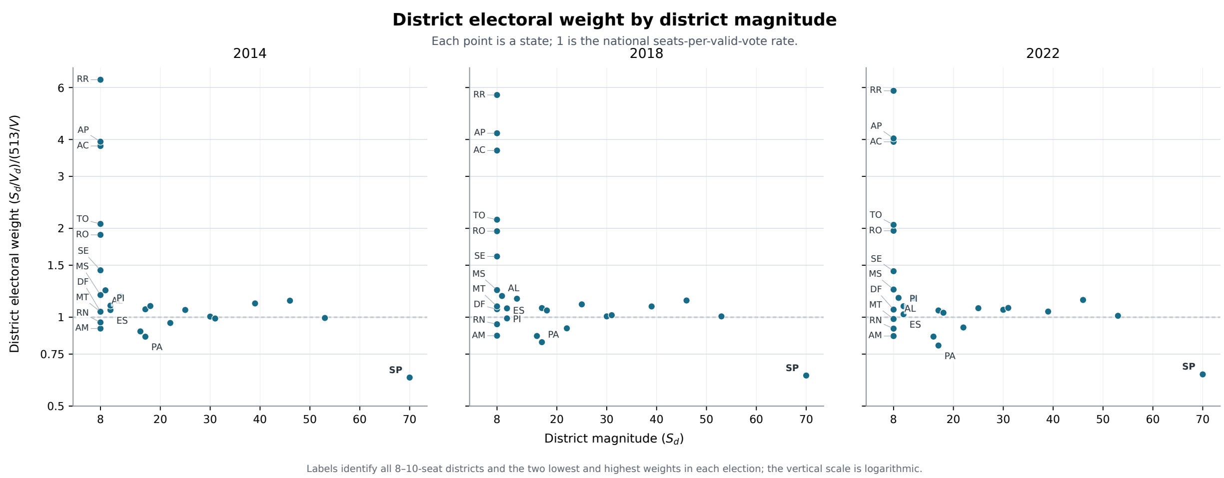

Figure 6: District electoral weight by district magnitude

Notes: Each point is a state or the Federal District. Electoral weight $w_d = 1$ is the national seats-per-valid-vote rate; the vertical scale is logarithmic. Labels identify all 8–10-seat districts and the two lowest and highest weights in each election. This is a measurement diagnostic, not an estimate of a district-magnitude effect or a coalition-specific contribution to B_C.

B Cabinet party sets and observation histories

Table 6: Distinct election-year cabinet party sets

Set	Election-year parties	Occ.	Days	Prov.	First observed	Last observed
14-01	PCdoB, PDT, PMDB, PP, PR, PRB, PROS, PSD, PT, PTB	1	77	0	2015-01-01	2015-03-18
14-02	PCdoB, PDT, PMDB, PP, PR, PRB, PSD, PT, PTB	1	365	0	2015-03-19	2016-03-17
14-03	PCdoB, PDT, PMDB, PP, PR, PROS, PSD, PT, PTB	1	13	0	2016-03-18	2016-03-30
14-04	PCdoB, PDT, PMDB, PP, PR, PSD, PT, PTB	1	14	0	2016-03-31	2016-04-13
14-05	PCdoB, PDT, PMDB, PR, PSD, PT, PTB	1	5	0	2016-04-14	2016-04-18
14-06	PCdoB, PDT, PMDB, PR, PT, PTB	1	23	0	2016-04-19	2016-05-11
14-07	DEM, PMDB, PP, PPS, PR, PRB, PSB, PSD, PSDB, PTB, PV	5	520	70	2016-05-12	2017-10-17
14-08	DEM, PMDB, PP, PPS, PR, PRB, PSD, PSDB, PTB, PV	5	158	7	2016-10-10	2018-03-20
14-09	DEM, PMDB, PP, PR, PRB, PSD, PSDB, PTB, PV	1	12	0	2018-03-21	2018-04-01
14-10	DEM, PMDB, PP, PRB, PSD, PSDB, PTB, PV	1	4	0	2018-04-02	2018-04-05
14-11	PMDB, PP, PRB, PSD, PSDB, PTB, PV	1	90	0	2018-04-06	2018-07-04
14-12	PMDB, PP, PRB, PSD, PSDB, PV	1	180	1	2018-07-05	2018-12-31
18-01	DEM, MDB, NOVO, PP, PSL	1	406	0	2019-01-01	2020-02-10
18-02	DEM, MDB, NOVO, PP, PSDB, PSL	2	8	4	2020-02-11	2020-04-16
18-03	DEM, NOVO, PP, PSDB, PSL	2	78	5	2020-02-18	2020-05-06
18-04	DEM, PP, PSDB, PSL	1	41	0	2020-05-07	2020-06-16
18-05	DEM, PP, PSD, PSDB, PSL	1	1	0	2020-06-17	2020-06-17
18-06	DEM, PP, PSD, PSL	1	93	0	2020-06-18	2020-09-18
18-07	DEM, PSD, PSL	1	81	0	2020-09-19	2020-12-08
18-08	DEM, PSC, PSD, PSL	1	65	0	2020-12-09	2021-02-11
18-09	DEM, PRB, PSC, PSD, PSL	1	47	1	2021-02-12	2021-03-30
18-10	DEM, PR, PRB, PSC, PSD, PSL	1	119	1	2021-03-31	2021-07-27
18-11	DEM, PP, PR, PRB, PSC, PSD, PSL	1	239	0	2021-07-28	2022-03-23
18-12	DEM, PP, PR, PRB, PSC, PSL	1	3	0	2022-03-24	2022-03-26
18-13	PP, PR, PSC	1	1	0	2022-03-27	2022-03-27
18-14	PP, PR, PRB, PSC	1	2	0	2022-03-28	2022-03-29
18-15	PP, PR, PRB	1	1	0	2022-03-30	2022-03-30
18-16	PP, PRB, PSD	1	41	0	2022-03-31	2022-05-10
18-17	DEM, PP, PRB, PSD, PSL	1	233	9	2022-05-11	2022-12-29
18-18	DEM, NOVO, PRB, PSD, PSL	1	2	2	2022-12-30	2022-12-31
22-01	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO	1	255	0	2023-01-01	2023-09-12
22-02	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	5	807	0	2023-09-13	2025-12-07
22-03	MDB, PCdoB, PDT, PP, PSB, PSD, PSDB, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	2	5	0	2023-12-04	2024-12-05
22-04	MDB, PCdoB, PDT, PP, PSB, PSD, PSDB, PSOL, PT, REDE, REPUBLICANOS	3	9	0	2025-10-08	2025-12-22
22-05	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS	2	98	0	2025-12-08	2026-03-19

Labels begin with the election year (14, 18, or 22) and order sets by first appearance within that election. Occ. counts separate appearances; Days sums their actual covered dates. First and last observed dates do not imply continuous observation. Prov. counts provisional days under the unchanged no-additional-party assumptions. Full intervals, historical periods, administrations, and daily evidence links are provided in the replication CSVs.

C Exact-connected minimal winning ideological intervals

Table 7 reports all minimal connected winning intervals in the primary $k = 0$ domain of Chamber-represented parties. These are contiguous intervals that reach the Chamber majority threshold and cease to do so when either endpoint is removed. The table is broader than Table 2, which reports only minimal connected interval inversions. The distinction matters because a minimal connected winning interval may have both a vote majority and a seat majority, while an interval inversion has a seat majority without a vote majority.

Table 7: Minimal connected winning ideological intervals

Election	Start	End	Parties	Vote %	Seats	Seat diff.	Status	Contains PT	Interval
2014	PCdoB	PMDB	12	50.17	265	7.63	votes+seats	Yes	PCdoB, PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB
2014	PT	PSD	12	54.34	291	12.26	votes+seats	Yes	PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD
2014	PDT	PSDB	12	51.80	276	10.25	votes+seats	No	PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB
2014	PSB	PTN	12	48.92	260	9.03	seats only	No	PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN
2014	PTB	PR	13	47.45	257	13.60	seats only	No	PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR
2014	PT DO B	PSDC	15	48.84	257	6.44	seats only	No	PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC
2014	SOLIDARIEDADE	PSL	15	48.84	257	6.46	seats only	No	SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL
2014	PMDB	PP	13	51.23	272	9.19	votes+seats	No	PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL, PP
2018	PT	PSDB	14	49.95	267	10.75	seats only	Yes	PT, PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB
2018	PDT	PR	18	50.59	268	8.47	votes+seats	No	PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR
2018	PSB	PRB	18	51.02	269	7.26	votes+seats	No	PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB
2018	SOLIDARIEDADE	PSL	15	50.12	262	4.86	votes+seats	No	SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL
2018	PMN	NOVO	15	50.93	257	-4.29	votes+seats	No	PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO
2018	MDB	PP	14	54.34	285	6.24	votes+seats	No	MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP
2018	PSD	PSC	14	50.55	258	-1.34	votes+seats	No	PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC
2018	PSDB	DEM	15	50.82	257	-3.72	votes+seats	No	PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC, PATRIOTA, DEM
2022	PT	PP	15	57.18	284	-9.36	votes+seats	Yes	PT, PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PSDB, PSD, PODE, PROS, PTB, PP
2022	PSB	PSC	16	53.82	261	-15.10	votes+seats	No	PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PSDB, PSD, PODE, PROS, PTB, PP, REPUBLICANOS, PSC
2022	MDB	UNIÃO	10	49.22	265	12.51	seats only	No	MDB, PSDB, PSD, PODE, PROS, PTB, PP, REPUBLICANOS, PSC, UNIÃO
2022	PP	PL	7	45.26	258	25.81	seats only	No	PP, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, NOVO, PL

D Robustness of the ideological coalition domain

Exact connectedness among seat-winning parties is the substantive baseline. The following checks relax connectedness and then vary the ideological party universe, retaining all valid federal-deputy votes in the denominator.

D.1 One-gap relaxation

The one-gap relaxation uses the same election-year order $p_1, \ldots, p_n$ of represented parties P^+ as the exact-connected baseline. For any nonempty $C \subseteq P^+$, let $a = \min\{j : p_j \in C\}$ and $b = \max\{j : p_j \in C\}$. Its span is $\mathrm{span}(C) = \{p_a, \ldots, p_b\}$, containing all represented parties between its leftmost and rightmost members. Define

$$g(C) = |\mathrm{span}(C) \setminus C|, \qquad \mathcal{D}_k = \{C \subseteq P^+ : C \neq \varnothing,\ g(C) \leq k\}.$$

Thus, $\mathcal{D}_0 = \{C \subseteq P^+ : C \neq \varnothing,\ g(C) = 0\}$ is the exact-connected domain, while $\mathcal{D}_1 = \{C \subseteq P^+ : C \neq \varnothing,\ g(C) \leq 1\}$ also permits one represented party inside the span to be omitted. The domains are nested, $\mathcal{D}_0 \subset \mathcal{D}_1$. A party with no seats contributes no gap in either primary domain. In each $\mathcal{D}_k$, a winning coalition is minimal if no proper subset that is also in $\mathcal{D}_k$ has at least 257 seats. Minimality is recomputed separately for $k = 0$ and $k = 1$; changing admissibility can change which coalitions are minimal. Inversions are then identified among these minimal winning coalitions using $v_C/V < 1/2$, with the same all-valid-vote denominator.

Table 8 compares the complete minimal-majority and minimal-inversion counts and the strongest inversion in each election and domain. The full coalition enumeration, including membership, omitted parties, votes, seats, minimality, and decomposition terms, is retained in the machine-readable replication outputs.

Table 8: Exact connectedness and the one-gap relaxation: minimal winning parliamentary coalitions

Election	k	Minimal majorities	Inversions	Strongest minimal inversion
2014	0	8	4	PTB–PR (47.45%; 257 seats)
2014	1	87	46	PTB–PR (47.45%; 257 seats)
2018	0	8	1	PT–PSDB (49.95%; 267 seats)
2018	1	108	42	PCdoB–PODE, omitting PSDB (47.59%; 257 seats)
2022	0	4	2	PP–PL (45.26%; 258 seats)
2022	1	39	12	PP–PL (45.26%; 258 seats)

Notes: The ideological order contains seat-winning parties. k is the maximum number of omitted represented parties inside a coalition's ideological span: $k = 0$ requires exact connectedness, and $k = 1$ also permits one omission. Minimality is defined within each stated domain. Vote shares use all valid federal-deputy votes. The strongest minimal inversion has the lowest national vote share; exact ties use coalition size and then canonical membership order.

Allowing one party inside a span to be omitted changes both admissibility and minimality. All twenty exact-connected minimal majorities remain admissible and winning, but only six remain minimal in $\mathcal{D}_1$. The one-gap domain contains 46 minimal inversions among 87 minimal majorities in 2014, 42 among 108 in 2018, and 12 among 39 in 2022. These are overlapping potential configurations under different restrictions, not probabilities of government formation.

In 2014, the relaxation broadens the ideological distribution of inversions: their leftmost members range from PSOL through PMDB, and seven include PT. The strongest inversion remains the same thirteen-party PTB–PR coalition, with 47.45 percent of the vote and 257 seats. Since every coalition member has positive seats, this exact-threshold coalition has no proper winning subset and remains minimal when one gap is allowed.

The strongest 2018 one-gap case is PCdoB–PODE omitting PSDB. The full span contains sixteen parties and holds 287 seats with 53.59 percent of all valid votes. Omitting PSDB removes thirty seats, leaving fifteen parties with exactly 257 seats and 47.59 percent of the vote. Because PSDB is underrepresented, this exclusion raises the differential by 0.76 seats, from 12.10 to 12.86. The coalition retains its seat majority while losing its vote majority. This exclusion leaves the coalition at the seat-majority threshold. The one-gap search thus yields a larger vote deficit than the PT–PSDB exact-connected inversion.

In 2022, all twelve one-gap minimal inversions contain only parties classified as center-right, right, or far right, and none includes PT. They contain seven to eleven parties, with a median of nine. The strongest remains the same seven-party PP–PL interval as in $k = 0$, with 45.26 percent of the vote and 258 seats. Three other minimal inversions occupy the wider MDB–UNIÃO span with one party omitted. The right-side concentration therefore extends beyond the single coalition with the lowest vote share.

Under the one-gap relaxation, three minimal inversions have a negative within-district component: two in 2014 and one in 2022. Their positive between-district terms more than offset that shortfall.

D.2 Sensitivity to the ideological party universe

The primary universe contains only parties winning Chamber seats, while the all-party sensitivity universe restores parties receiving valid votes but no seats to the ideological ordering. Both use all valid federal-deputy votes in the national denominator. Table 9 compares the number of parties, minimal winning coalitions, minimal inversions, and strongest cases for each election and admissibility condition. Complete coalition compositions, span and gap information, electoral metrics, and computed decomposition terms for both universes are supplied in the machine-readable replication outputs.

Table 9: Sensitivity to the ideological party universe

Election/k	Universe	$\|P^*\|$	MW	Inv.	Strongest inversion	Vote %	Seats
2014/0	Seat-winning	28	8	4	PTB–PR	47.45	257
2014/0	All parties	32	8	4	PTB–PR	47.59	257
2014/1	Seat-winning	28	87	46	PTB–PR	47.45	257
2014/1	All parties	32	88	43	PTB–PR, omitting PPL	47.45	257
2018/0	Seat-winning	30	8	1	PT–PSDB	49.95	267
2018/0	All parties	35	8	0	None	–	–
2018/1	Seat-winning	30	108	42	PCdoB–PODE, omitting PSDB	47.59	257
2018/1	All parties	35	118	24	PCdoB–PODE, omitting PSDB	47.82	257
2022/0	Seat-winning	23	4	2	PP–PL	45.26	258
2022/0	All parties	32	4	2	PP–PL	45.35	258
2022/1	Seat-winning	23	39	12	PP–PL	45.26	258
2022/1	All parties	32	53	17	PP–PL, omitting DC	45.26	258

Notes: $|P^*|$ is the number of parties in the specified ideological order; MW counts domain-minimal seat-majority coalitions, and Inv. counts inversions among them. Both universes use all valid federal-deputy votes in V. Strongest means the smallest national vote share, with ties resolved by membership count and coalition identifier. The strongest endpoint region changes for election/k = 2018/0; other comparisons preserve it.

The primary orders contain 28, 30, and 23 parties in 2014, 2018, and 2022; the all-party orders contain 32, 35, and 32. Exact-connected minimal-majority counts are unchanged, but the all-party universe has six minimal inversions instead of seven: its 2018 PT–PSDB interval includes the seatless PMB and has a vote majority. Under $k = 1$, the all-party inversion counts are 43, 24, and 17, compared with 46, 42, and 12 in the primary universe. The direction of the count difference therefore varies across elections.

The strongest 2014 and 2022 spans are PTB–PR and PP–PL in both universes. Their primary connected member sets coincide with the strongest all-party one-gap coalitions, which omit PPL and DC. In 2018, both one-gap searches select PCdoB–PODE omitting PSDB, but the primary coalition also excludes PMB by construction and receives 47.59 rather than 47.82 percent of all valid votes, with the same 257 seats. No vote denominator changes in these comparisons. Restoring extra-parliamentary parties changes which party sets are adjacent or admissible and which votes belong to coalition numerators.

The decomposition is sensitive in the 2022 MDB–UNIÃO case. Its primary within-district component is positive (2.95 seats), while the all-party component is negative (-1.29 seats); favorable between-district weighting supports inversion in both. The primary exact-connected inversions therefore all have positive within-district components, whereas five of the six all-party cases do. The existence of inversions and their strongest 2014 and 2022 locations survive, while the 2018 exact-connected result and this decomposition sign require the distinction between ideological universes.

Table 10 reports the all-party exact-connected minimal-inversion decomposition for direct comparison with Table 2.

Table 10: All-party sensitivity: exact-connected ($k = 0$) minimal inversions and seat-differential decomposition

Election	Start	End	Parties	Vote %	Seats	d_C	A_C	B_C
2014	PSB	PTN	12	48.92	260	9.032	10.509	-1.477
2014	PTB	PR	14	47.59	257	12.857	11.621	1.236
2014	PT DO B	PSDC	16	48.99	257	5.697	6.267	-0.570
2014	SOLIDARIEDADE	PSL	16	48.98	257	5.717	6.586	-0.869
2022	MDB	UNIÃO	15	49.98	265	8.591	-1.292	9.883
2022	PP	PL	8	45.35	258	25.355	22.821	2.534

E Cabinet coalitions and ideological intervals

All references to ideological intervals, connected closures, and minimal connected coalitions in this appendix section use the primary seat-winning universe and the exact-connected $k = 0$ baseline. The one-gap robustness results appear separately in Appendix D.1 and Table 8.

Table 11 compares each distinct election-year cabinet party set once with the ideological intervals that contain or approximate it. Cabinet membership is translated before grouping, and the linked chronology remains available separately. The connected closure is the smallest no-gap interval in the filtered parliamentary order containing the set's represented election-year parties. Gaps count represented parties inside that closure that are not cabinet members. Cabinet votes and seats sum the full translated set; the parliamentary restriction applies only to the ideological diagnostics. The final columns report the closest minimal connected winning interval and, where one exists, the closest minimal connected inversion. The comparison shows whether realized cabinet party sets are compact ideological blocs or sparse selections from a wider ideological span.

Table 11: Cabinet party sets and ideological intervals

Election	Set	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2014	14-01	votes+seats; 59.72%; 329 seats	left: 2; center-left: 1; center-right: 1; right: 6	PCdoB–PP	14	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PMDB–PP; 51.23%; 272 seats; overlap 6; J=0.35	PTB–PR; 47.45%; 257 seats; overlap 5; J=0.28
2014	14-02	votes+seats; 57.69%; 318 seats	left: 2; center-left: 1; center-right: 1; right: 5	PCdoB–PP	15	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.31	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.22
2014	14-03	votes+seats; 55.17%; 308 seats	left: 2; center-left: 1; center-right: 1; right: 5	PCdoB–PP	15	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.31	PTB–PR; 47.45%; 257 seats; overlap 5; J=0.29
2014	14-04	votes+seats; 53.14%; 297 seats	left: 2; center-left: 1; center-right: 1; right: 4	PCdoB–PP	16	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.33	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.24
2014	14-05	seats only; 46.54%; 259 seats	left: 2; center-left: 1; center-right: 1; right: 3	PCdoB–PR	12	left: 2; center-left: 2; center: 2; center-right: 5; right: 8	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.36	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.25
2014	14-06	neither; 40.41%; 223 seats	left: 2; center-left: 1; center-right: 1; right: 2	PCdoB–PR	13	left: 2; center-left: 2; center: 2; center-right: 5; right: 8	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.38	PTB–PR; 47.45%; 257 seats; overlap 3; J=0.19
2014	14-07	votes+seats; 64.27%; 346 seats	center-left: 1; center: 2; center-right: 1; right: 6; far right: 1	PSB–DEM	13	center-left: 1; center: 2; center-right: 5; right: 14; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.44	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.44
2014	14-08	votes+seats; 57.83%; 312 seats	center: 2; center-right: 1; right: 6; far right: 1	PPS–DEM	13	center: 2; center-right: 5; right: 14; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.38	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.38
2014	14-09	votes+seats; 55.82%; 302 seats	center: 1; center-right: 1; right: 6; far right: 1	PV–DEM	13	center: 1; center-right: 5; right: 14; far right: 2	PMDB–PP; 51.23%; 272 seats; overlap 6; J=0.38	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.31
2014	14-10	votes+seats; 50.03%; 268 seats	center: 1; center-right: 1; right: 5; far right: 1	PV–DEM	14	center: 1; center-right: 5; right: 14; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.33	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.33
2014	14-11	neither; 45.83%; 247 seats	center: 1; center-right: 1; right: 5	PV–PP	12	center: 1; center-right: 5; right: 13	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.36	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.36
2014	14-12	neither; 41.81%; 222 seats	center: 1; right: 5	PV–PP	13	center: 1; center-right: 5; right: 13	PMDB–PP; 51.23%; 272 seats; overlap 5; J=0.36	PSB–PTN; 48.92%; 260 seats; overlap 4; J=0.29
2018	18-01	neither; 30.14%; 160 seats	right: 4; far right: 1	MDB–DEM	12	right: 15; far right: 2	MDB–PP; 54.34%; 285 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-02	neither; 36.13%; 190 seats	right: 5; far right: 1	MDB–DEM	11	right: 15; far right: 2	MDB–PP; 54.34%; 285 seats; overlap 5; J=0.33	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.11
2018	18-03	neither; 30.60%; 156 seats	right: 4; far right: 1	PSDB–DEM	10	right: 13; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 5; J=0.33	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06

Election	Set	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2018	18-04	neither; 27.80%; 148 seats	right: 3; far right: 1	PSDB–DEM	11	right: 13; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-05	neither; 33.65%; 183 seats	right: 4; far right: 1	PSD–DEM	11	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.12
2018	18-06	neither; 27.65%; 153 seats	right: 3; far right: 1	PSD–DEM	12	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 3; J=0.20	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-07	neither; 22.15%; 116 seats	right: 2; far right: 1	PSD–DEM	13	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 2; J=0.13	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-08	neither; 23.90%; 123 seats	right: 3; far right: 1	PSD–DEM	12	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 3; J=0.20	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-09	neither; 28.97%; 152 seats	right: 4; far right: 1	PSD–DEM	11	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-10	neither; 34.29%; 185 seats	right: 5; far right: 1	PSD–DEM	10	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 5; J=0.33	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05
2018	18-11	neither; 39.79%; 222 seats	right: 6; far right: 1	PSD–DEM	9	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 6; J=0.40	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05
2018	18-12	neither; 33.94%; 187 seats	right: 5; far right: 1	PR–DEM	4	right: 8; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 6; J=0.40	PT–PSDB; 49.95%; 267 seats; overlap 0; J=0.00
2018	18-13	neither; 12.57%; 77 seats	right: 3	PR–PSC	5	right: 8	PSD–PSC; 50.55%; 258 seats; overlap 3; J=0.21	PT–PSDB; 49.95%; 267 seats; overlap 0; J=0.00
2018	18-14	neither; 17.64%; 106 seats	right: 4	PR–PSC	4	right: 8	PSD–PSC; 50.55%; 258 seats; overlap 4; J=0.29	PT–PSDB; 49.95%; 267 seats; overlap 0; J=0.00
2018	18-15	neither; 15.89%; 99 seats	right: 3	PR–PP	4	right: 7	PSD–PSC; 50.55%; 258 seats; overlap 3; J=0.21	PT–PSDB; 49.95%; 267 seats; overlap 0; J=0.00
2018	18-16	neither; 16.43%; 101 seats	right: 3	PSD–PP	10	right: 13	PSD–PSC; 50.55%; 258 seats; overlap 3; J=0.21	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-17	neither; 32.73%; 182 seats	right: 4; far right: 1	PSD–DEM	11	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06
2018	18-18	neither; 30.02%; 153 seats	right: 4; far right: 1	PSD–DEM	11	right: 14; far right: 2	PSD–PSC; 50.55%; 258 seats; overlap 4; J=0.27	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.06

Election	Set	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2022	22-01	seats only; 48.89%; 263 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 1; far right: 1	PSOL–UNIÃO	11	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 3	PT–PP; 57.18%; 284 seats; overlap 6; J=0.33	MDB–UNIÃO; 49.22%; 265 seats; overlap 3; J=0.19
2022	22-02	votes+seats; 63.79%; 350 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 2	PSOL–UNIÃO	9	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 3	PT–PP; 57.18%; 284 seats; overlap 7; J=0.37	MDB–UNIÃO; 49.22%; 265 seats; overlap 5; J=0.31
2022	22-03	votes+seats; 66.81%; 363 seats	far left: 1; left: 2; center-left: 3; center-right: 2; right: 2; far right: 2	PSOL–UNIÃO	8	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 3	PT–PP; 57.18%; 284 seats; overlap 8; J=0.42	MDB–UNIÃO; 49.22%; 265 seats; overlap 6; J=0.38
2022	22-04	votes+seats; 57.48%; 304 seats	far left: 1; left: 2; center-left: 3; center-right: 2; right: 2; far right: 1	PSOL–REPUBLICANOS	7	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 1	PT–PP; 57.18%; 284 seats; overlap 8; J=0.44	MDB–UNIÃO; 49.22%; 265 seats; overlap 5; J=0.31
2022	22-05	votes+seats; 54.45%; 291 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 1	PSOL–REPUBLICANOS	8	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 1	PT–PP; 57.18%; 284 seats; overlap 7; J=0.39	MDB–UNIÃO; 49.22%; 265 seats; overlap 4; J=0.25

One row per distinct election-year cabinet party set. Set labels refer to Appendix B; full occurrence linkage is retained in the replication CSVs.

The first result is that no observed cabinet coalition is itself a contiguous ideological interval. Every cabinet party set has gaps inside its connected closure. The number of omitted represented parties ranges from four to sixteen. Brazilian cabinet coalitions are therefore selective combinations within wider parliamentary ideological spans, omitting many represented parties between their leftmost and rightmost members.

This matters for the interpretation of cabinet inversions. The observed cabinet inversions are sparse coalitions, but their connected closures are not inversions. All connected closures in Table 11 are vote-and-seat majorities. Cabinet party set 14-05, for example, has 46.54 percent of the vote and 259 seats, but its connected closure runs from PCdoB to PR and contains 19 represented parties. That closure has 77.46 percent of the vote and 408 seats. The inversion therefore does not come from a compact left-to-right ideological bloc. It comes from a selective cabinet coalition that retains a seat majority while omitting twelve represented parties inside its ideological span.

The first Temer cabinet party set, 14-07, contains DEM, PMDB, PP, PPS, PR, PRB, PSB, PSD, PSDB, PTB, and PV. It recurs on five occasions, covering 520 days in total. Its connected closure runs from PSB to DEM, contains 24 represented parties, and has thirteen gaps. The cabinet has 64.27 percent of the vote and 346 seats, while the closure has 78.27 percent and 409 seats. Neither is an inversion. Its closest minimal connected winning interval, PSB–PTN, is an inversion with 48.92 percent of the vote and 260 seats. This places a realized vote-and-seat majority near a potential connected inversion without conflating the two memberships.

The Dilma cabinet party sets have a different ideological profile from the later Temer sets. They are sparse but left-anchored, including PT, PCdoB, and PDT while extending to parties such as PMDB, PSD, PR, PP, and PTB. Their closures run from PCdoB to PR or PP. The transition on May 12, 2016 changes this structure: from set 14-06 to 14-07, PCdoB, PDT, and PT leave, while DEM, PP, PPS, PRB, PSB, PSD, PSDB, and PV enter. The cabinet remains non-contiguous, but its ideological anchor shifts rightward.

The ideology diagnostics confirm this dated transition. From set 14-06 to 14-07 on May 12, 2016, the unweighted cabinet ideology mean rises by 1.77 points and the seat-weighted mean by 1.65 points. These changes compare the memberships on either side of that transition, without weighting the comparison by later recurrences. The pre-Temer set is left-anchored and broad; the Temer set is center-to-right and broad. Both are non-contiguous, but they occupy different regions of the ideological order.

The Bolsonaro cabinet party sets are more concentrated on the right and far right. Set 18-11, observed from July 28, 2021 through March 23, 2022, has 39.79 percent of the vote and 222 seats. Its parliamentary closure, PSD–DEM, contains sixteen parties and nine gaps, with 56.67 percent of the vote and 292 seats. The cabinet is a minority in both measures, while its closure has both majorities. Its closest minimal connected winning interval is PSD–PSC, with 50.55 percent of the vote and 258 seats. The closest minimal connected inversion is PT–PSDB, with 49.95 percent and 267 seats, but its Jaccard overlap with the cabinet is only 0.050. The exact-connected inversion therefore occupies a different ideological region from this observed cabinet minority.

The Lula cabinet party set 22-01 returns to a left-anchored structure, but it is not a compact left coalition. It contains PSOL, PCdoB, PT, PDT, PSB, REDE, MDB, PSD, and UNIÃO. Its parliamentary closure runs from PSOL to UNIÃO, contains 20 represented parties, and has eleven gaps. The observed cabinet is an inversion, with 48.89 percent of the vote and 263 seats, but its connected closure has 79.83 percent of the vote and 407 seats. The closest minimal connected winning interval is PT–PP, a broad vote-and-seat majority with 57.18 percent of the vote and 284 seats. The closest minimal connected inversion is MDB–UNIÃO, which lies to the right of the cabinet's left anchor.

This comparison is crucial for interpreting the 2022 result. The Lula cabinet is a realized inversion, but it is not the compact connected inversion revealed by the interval search. The compact connected inversion is PP–PL: a seven-party right/far-right parliamentary interval with 45.26 percent of the vote and 258 seats. The observed Lula coalition reaches a seat majority through a broad and sparse cross-ideological cabinet arrangement. The exact-connected interval analysis shows that the most compact vote-minority seat-majority bloc in the same Chamber is located on the right.

The two analyses therefore meet at a specific structural point. Cabinet coalitions identify realized governing party sets. Connected intervals identify the seat-vote geometry of ideological blocs. The cabinet-to-interval comparison shows that realized cabinets may draw on seat asymmetries without being connected blocs themselves. In 2014, the pre-Temer coalitions are sparse and left-anchored, while the Temer coalitions are sparse and shifted rightward. In 2018, set 18-11 is a minority in both votes and seats, while its connected closure has both majorities. In 2022, the Lula cabinet is a broad realized inversion, while the compact connected inversion lies on the right. The gap between cabinet coalitions and connected intervals is therefore not a nuisance. It shows how Brazil's coalition presidentialism combines sparse cabinet construction with an ideological distribution in which seat-efficient connected blocs are often located elsewhere.